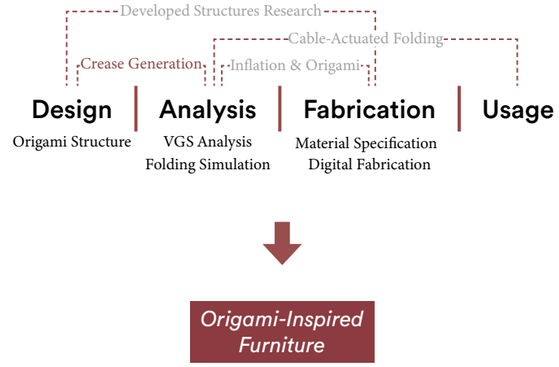
OriTriO

Origami-Inspired Triple Furniture from One Sheet

Academic Project

This project examines how a single-sheet origami system can be computationally derived to support three structurally stable and functionally distinct furniture morphologies. We developed an integrated workflow that combines behavior-informed geometric analysis, crease-pattern derivation, multi-stage crease integration, folding kinematic evaluation, and vector-based graphic statics. The process incorporates data-driven behavioral mapping, systematic crease consolidation, and VGS-based performance validation to couple motion feasibility with structural reliability. The resulting prototype demonstrates a fully encoded crease system capable of guiding reliable transitions among stool, chair, and bench configurations. This work advances a scalable methodology for high-performance origami structures, showing how computational geometry and material logic can enable adaptive, sustainable, and deployable design. I am motivated to develop performance-driven folding workflows that align computational design with responsive structural behavior.

Keywords: Origami, Transformative Structure, VGS
Instructor: Philip F. Yuan, Hua Chai, Yige Liu
Role: Origami Simulation + VGS Analysis + Digital Fabrication
Team: Suming Li, Ueki Wong, Yunning Feng
Contribution: Formative Study, Crease Design, Folding Simulation, VGS Analysis, Construction
Year: 2025



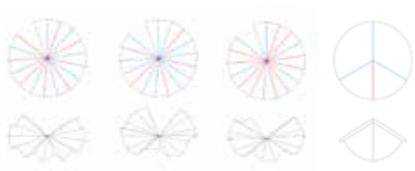
Workflow | *A Pipeline from Observation to Fabrication*

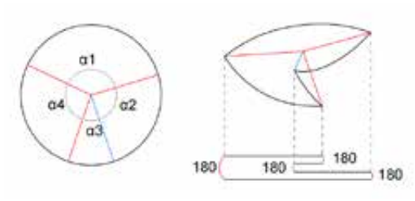
This page outlines the comprehensive workflow for the multi-form folding chair, from conceptual research to physical realization. The process begins with ergonomic mapping, advances through computational form-finding and iterative refinement, details key structural nodes, and culminates in precision fabrication, translating dynamic transformation into a tangible object.

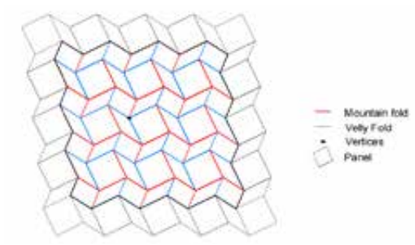
Design Research

Establishes design objectives through origami studies and site analysis. The findings provide clear direction for subsequent development.

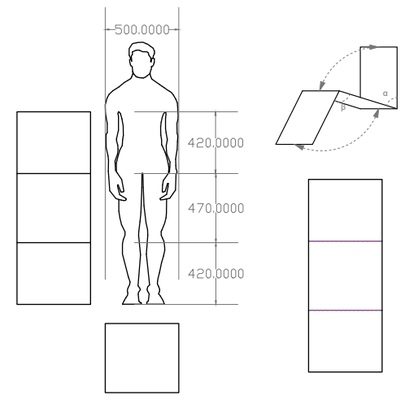
Origami Research







Ergonomics



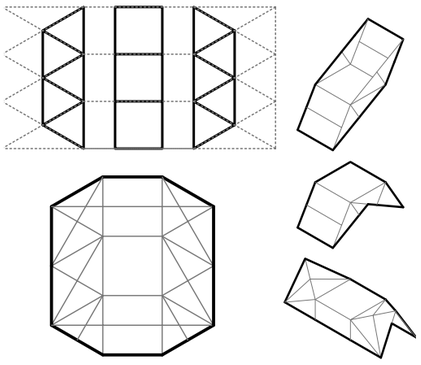
Crease Iteration

Iterates crease patterns using physical models and computational simulation. The process balances structural integrity with functional requirements.

Transformative Folding

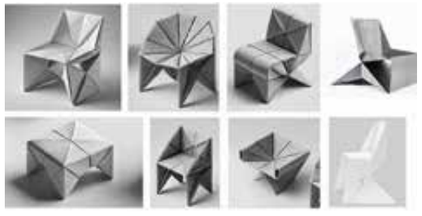
Multi-function

Grid Alignment

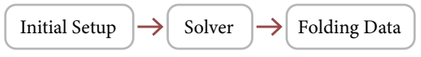


AI Generation

AI generation as the reference



Origami Simulation

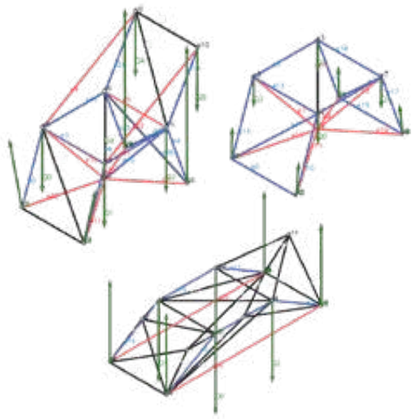


Folding Analysis

Employs Vector-based Graphic Statics for structural evaluation. Optimizes performance through stabilized unit implementation.

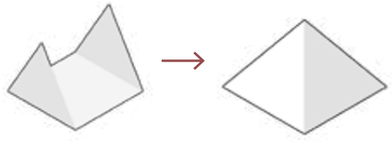
VGS

Structure Analysis



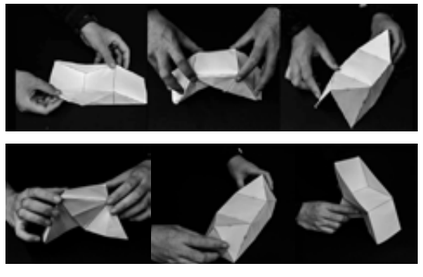
Optimalization

Structure Unit



Kinematic Verification

Physical Origami Prototyping

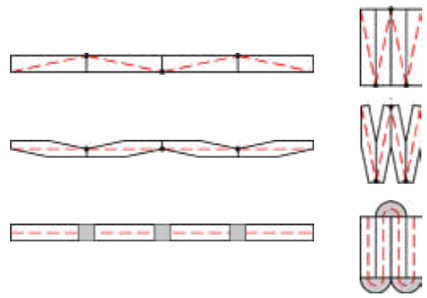


Joint Detailing

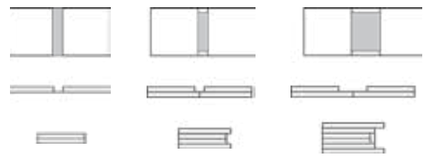
Develops hinge mechanisms and selects panel materials. Ensures connections accommodate movement while maintaining durability.

Hinge Types

Thick Fold



Full Fold



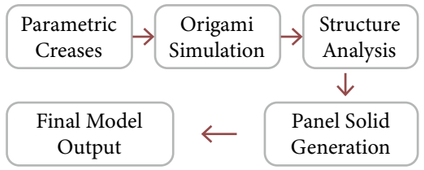
Panel Material

Strength | Density | Finish

Fabrication

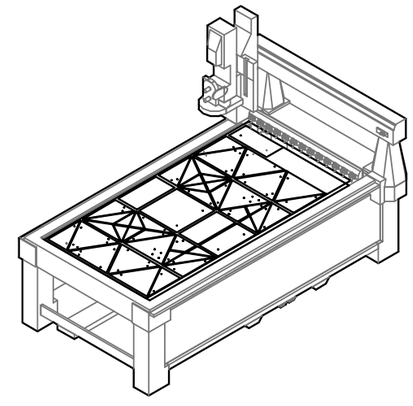
Integrates parametric processes from adjustment to analysis. Achieves precision through CNC manufacturing.

Digital Workflow

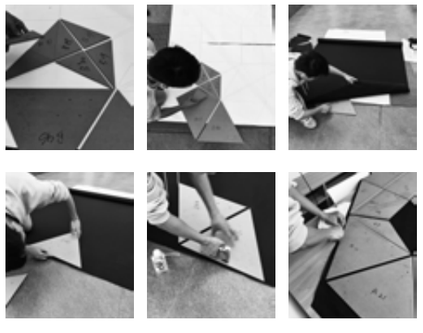


Intelligent Fabrication

CNC



Assembly



Foundational Research | *Mapping Site Dynamics and User Behaviors*

This section presents the genesis of our design. Rooted in a thorough site analysis, it articulates the spatial and functional advantages of employing a folding structure. Through systematic observation, we cataloged and synthesized predominant user activities, ultimately distilling key behavioral patterns that informed our primary design objectives.

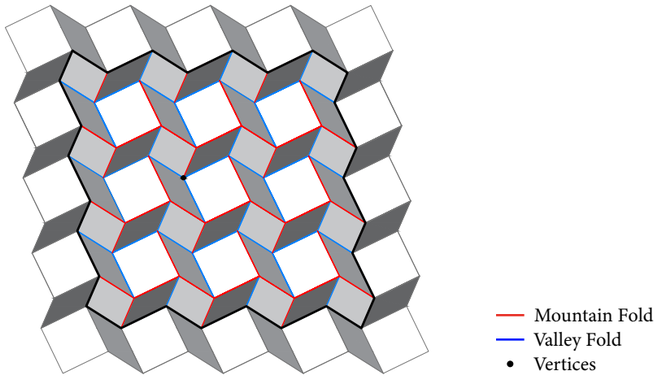
Origami Research

Structural Elements

An origami structure is locally a 2D discrete manifold composed of creases, which can be straight or curved and are defined by their folding angles. These creases divide the surface into panels and intersect at vertices, forming mountain or valley folds depending on direction and viewpoint.

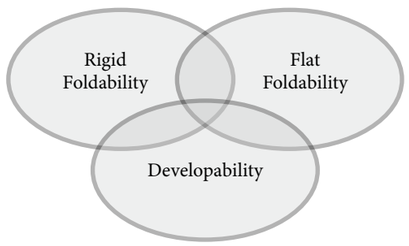
Maekawa-Justin Theorem

The difference between mountain (M) and valley (V) folds must be: $|M - V| = 2$



Core Properties

Origami has three core properties: developability, flat-foldability, and rigid foldability. Developability allows construction from a flat sheet; flat-foldability enables complete flattening; rigid foldability requires rigid panels moving only at creases. They are not always concurrent.



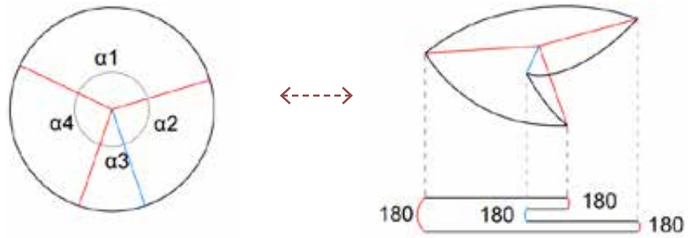
Developability Condition

A vertex is developableif the sum of its adjacent panel angles equals 2π: $\alpha_1 + \alpha_2 + \ldots + \alpha_n = 2\pi$

Flat-Foldability Condition

Kawasaki-Justin Theorem:

A vertex with an even number of creases is flat-foldable if: $\alpha_1 - \alpha_2 + \alpha_3 - \ldots - \alpha_n = 0$

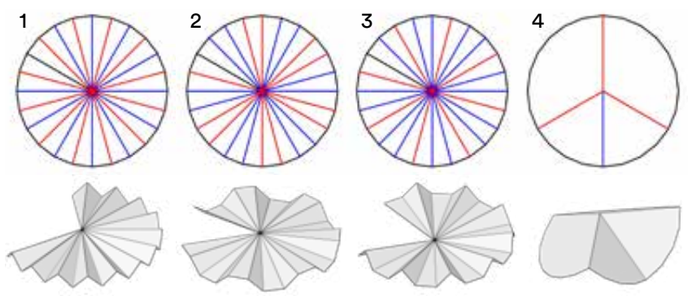


1. Accordion Fold:
Mountain and valley folds intersect alternately.

2. Door-Shaped Fold:
Two consecutive segments of the same fold type.

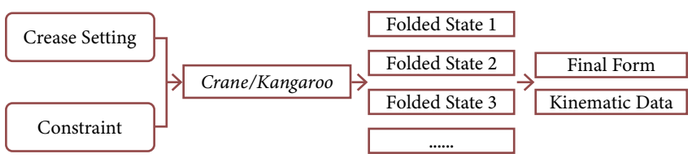
3. Tree-Like Fold: Two consecutive segments of the same fold type.

4. V-Shaped Fold:
A minimal node case where three valley (or mountain) folds converge into one mountain (or valley) fold.



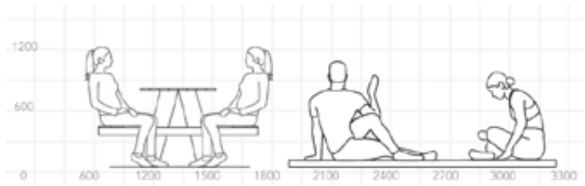
Computer Simulation Process

Folding behavior is defined in Grasshopper through creases and constraints, driven by origami mechanisms. Multiple states are sampled along the motion to obtain the final form and kinematic data.

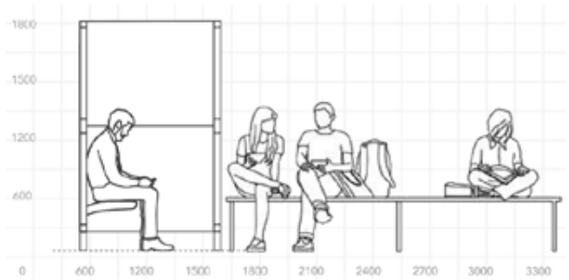


Behavioral Metrics

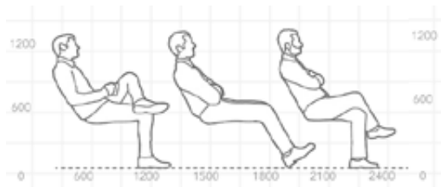
This section catalogs and documents observed behavioral patterns, translating them into quantifiable spatial dimensions and metrics that critically inform the design parameters.



Gathering



Waiting



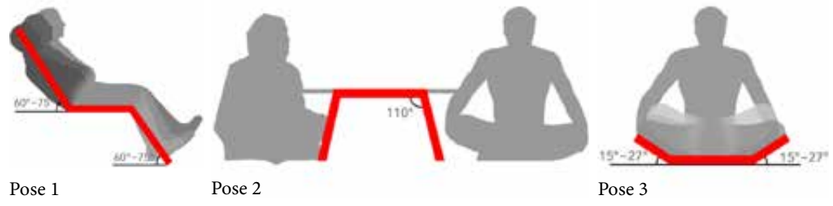
Draw

Meditation

Picnic

Pose Pattern

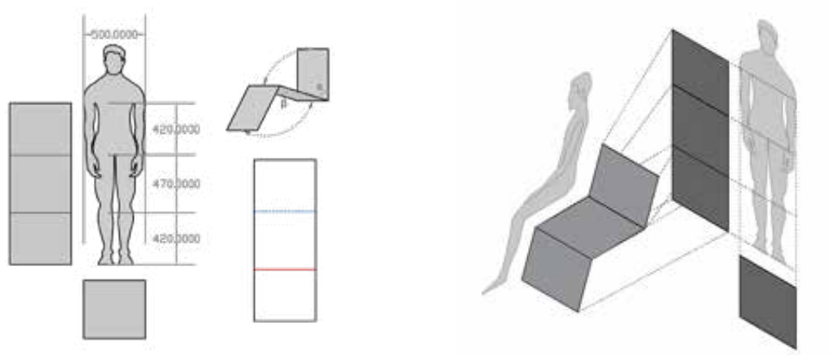
This section systematically analyzes and extracts key postures from behavioral studies, categorizing them into three primary pose typologies.



Pose 1 Pose 2 Pose 3

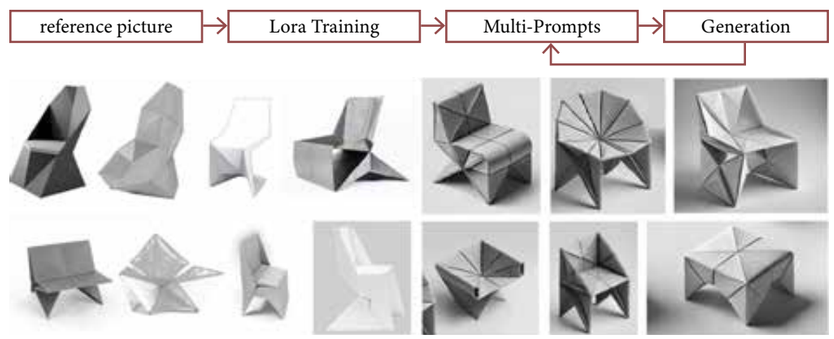
Posture–Form Mapping

Building on the prior extraction of posture types, this step translates human postures into dimensional constraints to generate primary grids and folding units. Posture-specific contact and load conditions are then used to define folding angle ranges, forming an operative link between behavior and geometry.



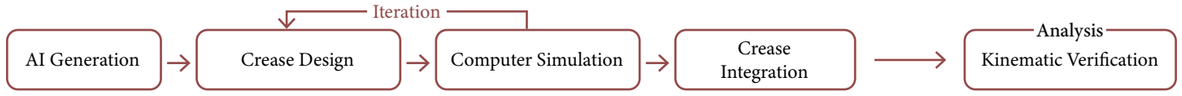
AI Generation

The AI-generated images served as preliminary visual experiments, offering broad morphological inspiration—particularly for the overall silhouette and support structure.

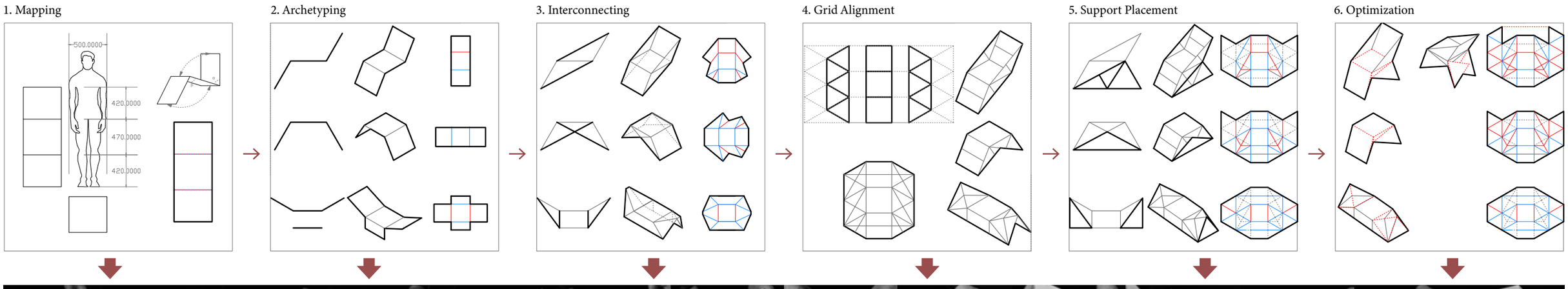


Form Finding | *Guiding Iteration Toward Design Goals*

This page captures the virtual exploration of the folding design. It begins with AI-generated possibilities, moves through key sequences of the folding simulation, and documents how insights from these models guided the refinement of the crease pattern, with physical prototypes serving as the ultimate test.

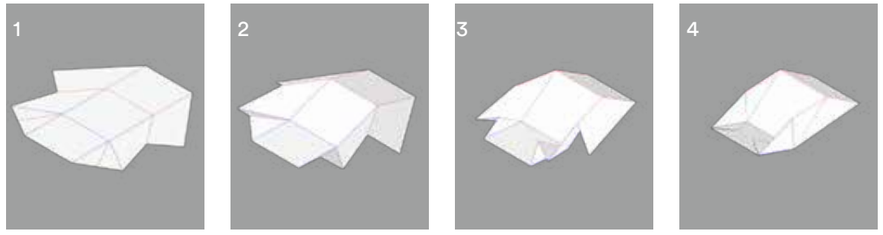


Iteration Pipeline

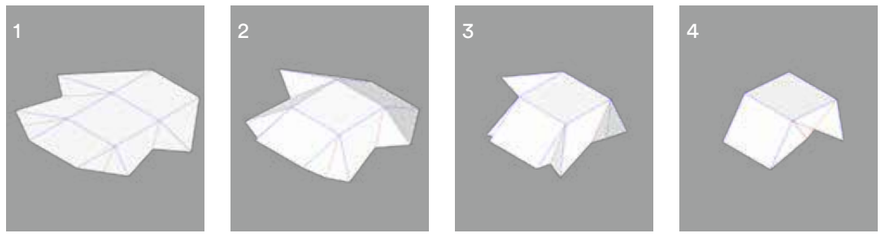


Computer Simulation

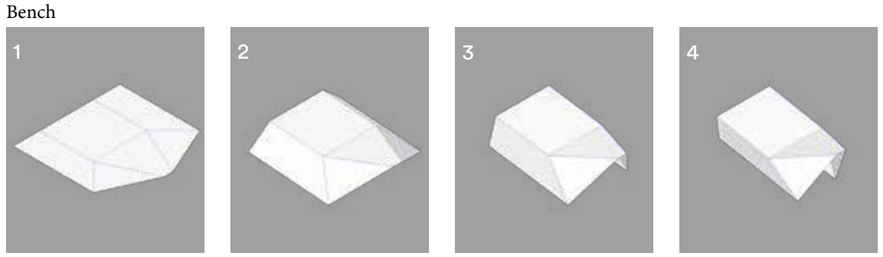
Chair



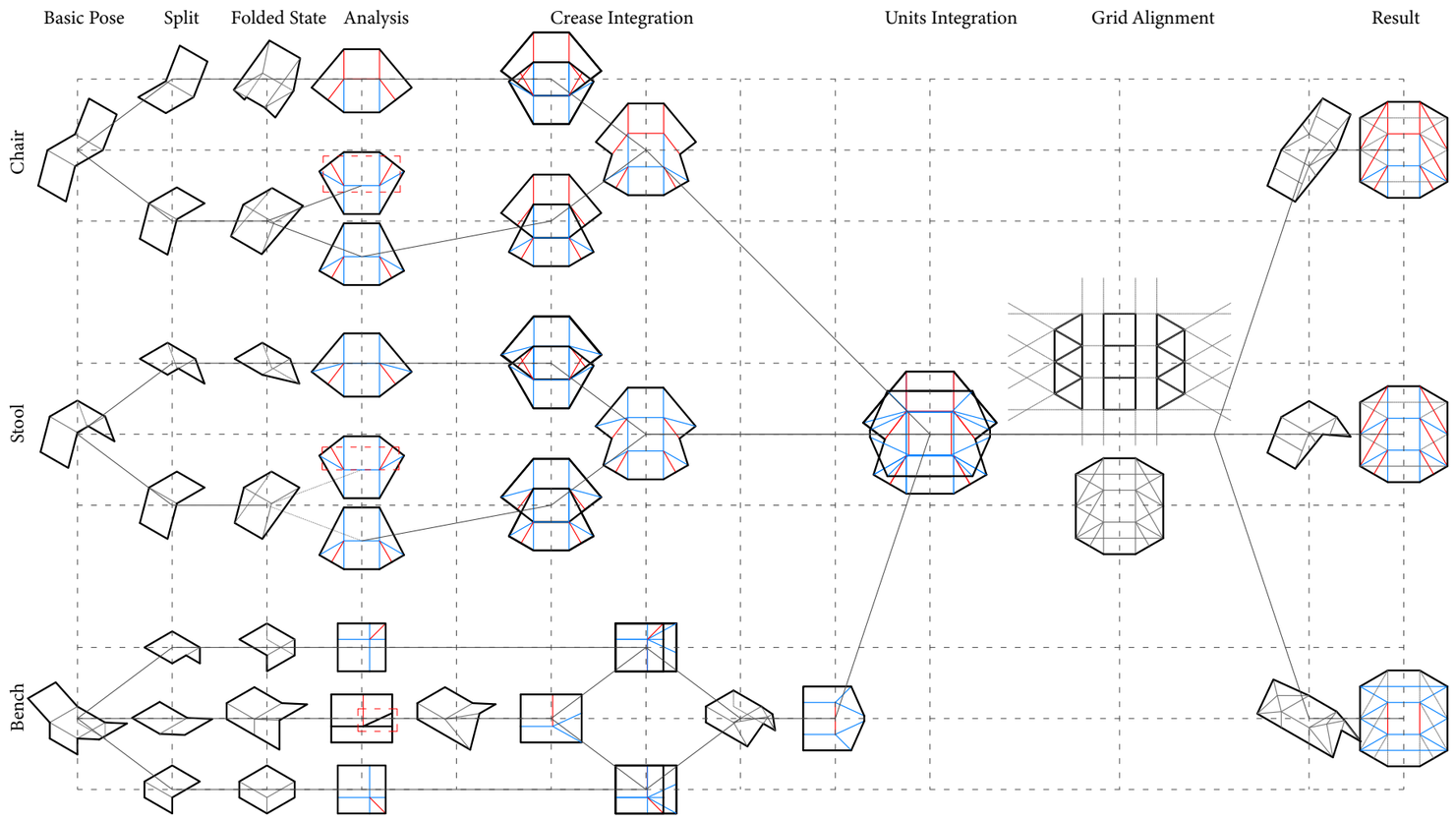
Stool



Bench



Iteration Process



Structural Fabrication | *From Analytical Insight to Constructive Precision*

This page presents the structural logic and fabrication workflow of the folding chair. Graphic statics revealed instability in the plate model, leading to the introduction of a closed geometric unit shared across all configurations. The fabrication section shows hinge studies, full folding thickness strategies, and CNC milling and assembly documentation.

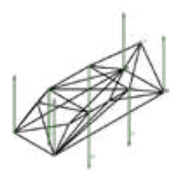
Structure Optimization

Original Form
Three furniture morphologies are defined as folded plate geometries to establish a common starting point for structural analysis.

Linkage
The plate geometries are abstracted into linkage systems of nodes and members for structural analysis.

Force Analysis
Graphic statics is applied to examine load paths and equilibrium conditions within the structure.

Initial Results
Initial analysis indicates kinematic instability, requiring additional constraints to achieve equilibrium.

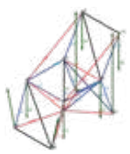
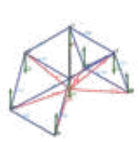
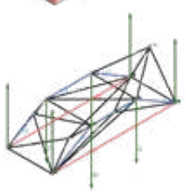
Chair	Stool	Bench
		
EQUILIBRIUM MATRIX 36x37 S (Self Stresses) = 5 M (Internal Mechanisms) = 4 The structure is statically indeterminate The structure is kinematically unstable: 4 bracing edges or support force variables have to be introduced	EQUILIBRIUM MATRIX 30x33 S (Self Stresses) = 5 M (Internal Mechanisms) = 2 The structure is statically indeterminate The structure is kinematically unstable: 2 bracing edges or support force variables have to be introduced	EQUILIBRIUM MATRIX 36x40 S (Self Stresses) = 4 M (Internal Mechanisms) = 0 The structure is statically indeterminate 4 self-stresses have to be assigned as parameters

Unit Optimization
A shared structural unit is introduced to optimize all configurations toward static determinacy.

Form Diagram
The form diagram abstracts geometric configuration into a structural framework.

Force Diagram
The force diagram represents internal force relationships corresponding to the form diagram.

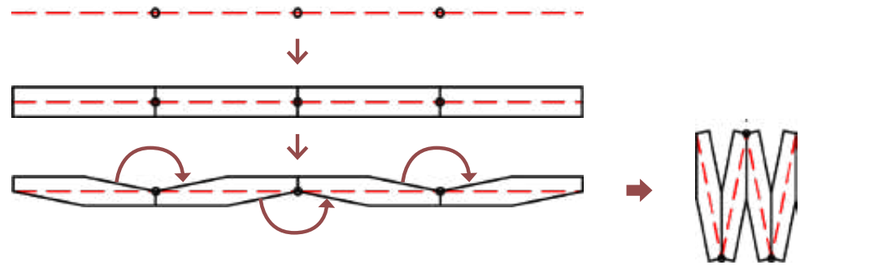
Final Result
The optimized structures achieve static determinacy with no internal mechanisms, validating the structural logic of the folding system.

		
EQUILIBRIUM MATRIX 33x35 S (Self Stresses) = 2 M (Internal Mechanisms) = 0 The structure is statically indeterminate 2 self-stresses have to be assigned as parameters	EQUILIBRIUM MATRIX 27x29 S (Self Stresses) = 2 M (Internal Mechanisms) = 0 The structure is statically indeterminate 2 self-stresses have to be assigned as parameters	EQUILIBRIUM MATRIX 36x40 S (Self Stresses) = 4 M (Internal Mechanisms) = 0 The structure is statically indeterminate 4 self-stresses have to be assigned as parameters

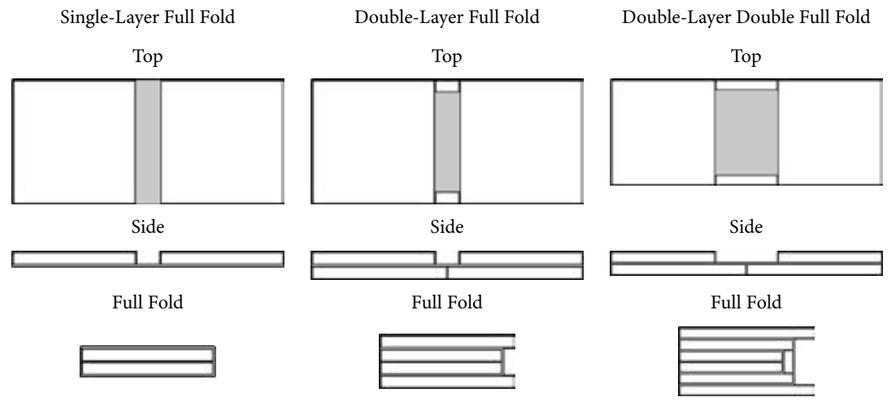
Hinge Research

Through research on thick-panel folding and full fold treatment, we selected the Bi-Groove Hinge. This hinge allows a single crease to achieve multiple folding modes, enabling dynamic morphological transformations.

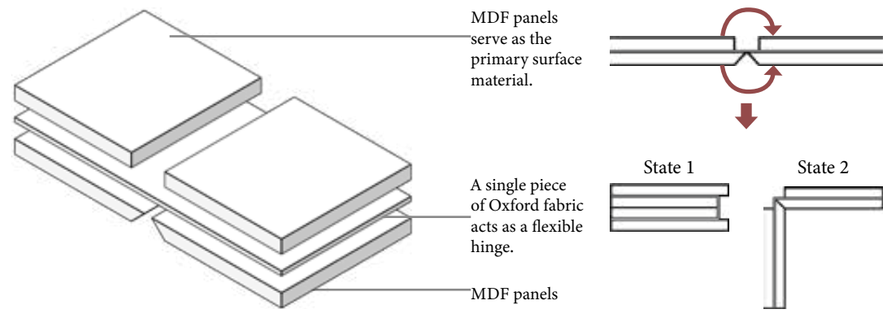
Thick Fold
Thick fold studies address the challenge of folding thick panels by introducing grooved cuts, allowing controlled rotation without material collision.



Full Fold
To accommodate fully collapsed configurations, gaps are strategically reserved to prevent overlap and enable complete folding.

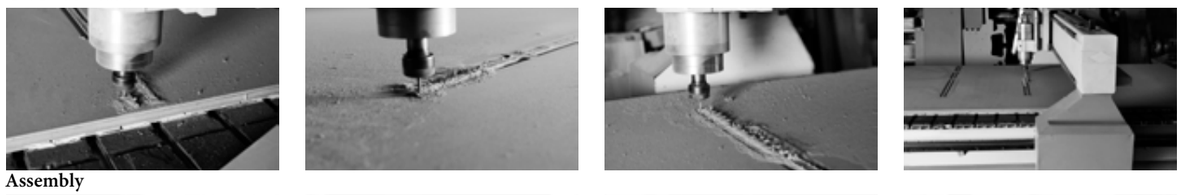


Bi-Groove Hinge
The bi-groove hinge enables folding along multiple directions, allowing a single crease system to support different furniture configurations.

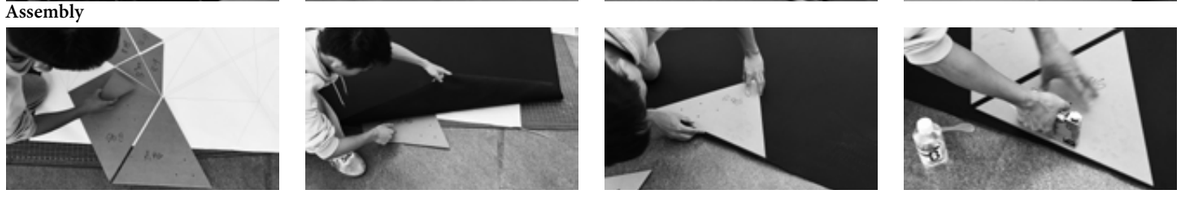


Construction

CNC Fabrication



Assembly



Final Realization | *One Sheet Forming Three Folded States*

This page presents the culmination of the design exploration. It dynamically illustrates the folding logic of the three morphologies and culminates in renderings and photographs, marking the successful translation of a complex structure from digital concept to physical reality.

Future Research | *Towards Lighter, Smarter, and More Responsive Structures*

This page outlines three pivotal directions for future research. The first explores pneumatic folding structures, integrating variable air-pressure systems with rigid panels to achieve an optimal balance between structural integrity and lightweight design. The second investigates tensile actuation systems, focusing on controlling nodal motion paths to enable single-action deployment. The final direction examines deployable structures, developing innovative methods for constructing complex curved surfaces from flat materials.

Inflation & Origami

This approach explores new morphological possibilities by integrating rigid folding with soft inflatable systems, creating hybrid rigid-soft structures.

Design | **Analysis** | **Fabrication** | Usage

INTERACTION

Path 1

Form → VGS → Tension / Compression → Inflation → Adaptive Structure Inflatable Material

Path 2

Form → VGS → Critical Structure → Rigid Panel

Form → VGS → Functional Panel → Hybrid Performance Structures Inflatable Material

Stability Transformability Portability

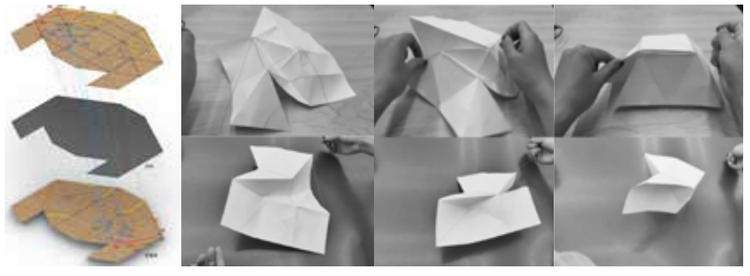
Cable-Actuated Folding

By analyzing origami vertex motion paths, strategic threading between key points enables controlled morphological switching.

Design | **Analysis** | Fabrication | **Usage**

INTERACTION

Form → Computer Simulation Folding Process → Vertex Trajectories / Crease Motions / Panel Kinematics → Cable-Actuated System



Rapid-Deployability

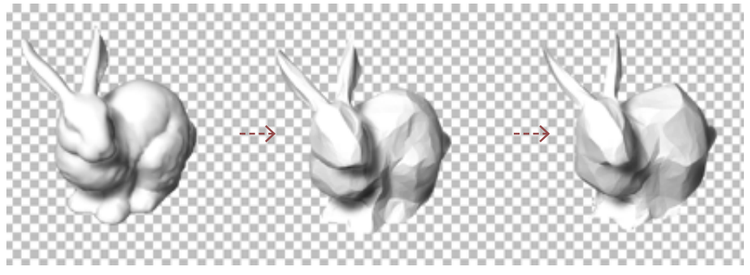
Developed Structures Research

This process optimizes complex forms into deployable structures that balance manufacturability, structural performance, and functional requirements.

Design | Analysis | **Fabrication** | Usage

INTERACTION

Form → Deployability Optimization → Motion Planning / Morphological Control / Fabrication Integration



Fabrication

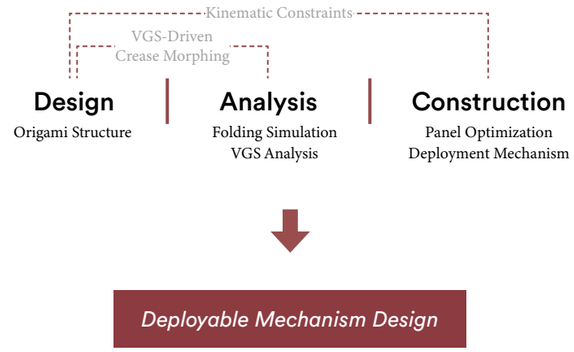
Umbrella Morphology

Vector-Based Graphic Statics in Deployable Architectural Design

Academic Project

This project establishes an integrated workflow that links origami geometry with deployable architectural performance. The framework combines parametric crease-pattern generation, folding-state computation, and vector-based structural evaluation through VGS. Its pipeline coordinates Kangaroo-based kinematic simulation, VGS-driven force–form analysis, and structurally informed geometric refinement. The final prototype demonstrates a coherent umbrella-type system whose geometry, load path, and construction details remain aligned throughout deployment. The research shows how embedding structural feedback into early geometric design enables adaptive, lightweight, and fabrication-ready foldable systems. This work reflects my ongoing interest in performance-driven and responsive origami structures.

Keywords: VGS, Foldable Structure, Responsive Structure
Instructor: Pierluigi D'Acunto, Philip F. Yuan, Patrick Ole Ohlbrock, Yuchi Shen, Junwang Yu
Role: Computational Origami + VGS Analysis + Digital Fabrication
Team: Suming Li, Ueki Wong, Yunning Feng
Contribution: Parametric Design, Folding Simulation, Structure Analysis, Details Design, Construction
Year: 2025



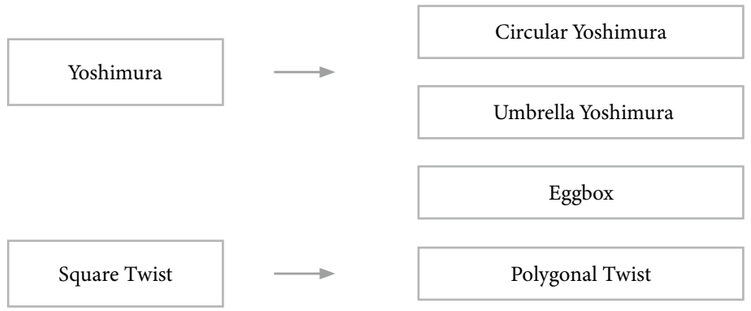
Workflow | *From Form-Finding to Physical Construction*

This workflow shows how origami-based geometry is transformed into an architectural deployable system. It begins with crease-pattern reinterpretation through parametric modeling and folding simulations, followed by kinematic and structural evaluation to understand motion behavior. Construction detailing and prototyping then translate the folding logic into a functional umbrella mechanism.

Design

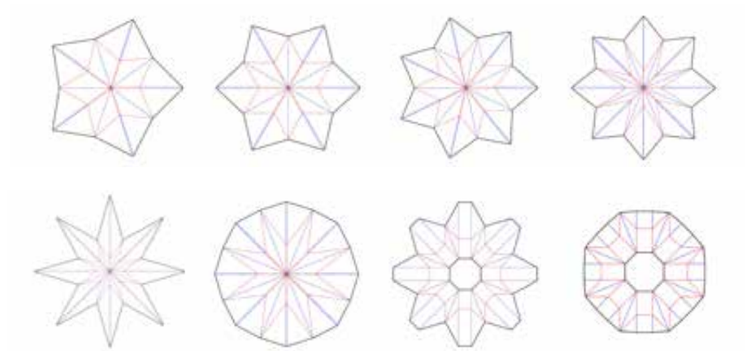
This phase explores how existing origami crease patterns can be reinterpreted into an architectural deployable system. Parametric modeling and digital folding simulations are used to iterate rapidly and test the geometric logic of the mechanism.

Origami Patterns

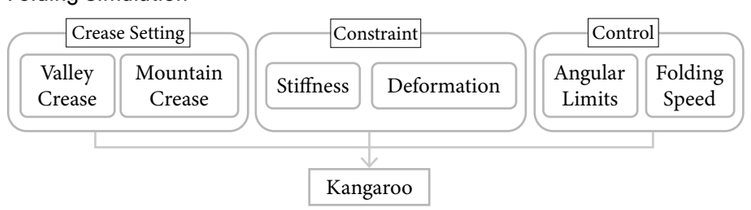


Parametric Transformation

Unit Extraction → Parametric Tessellation



Folding Simulation



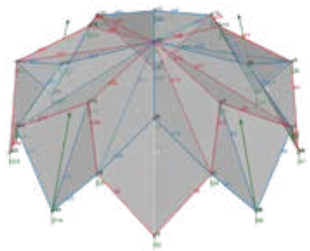
Analysis

The motion behavior of the folding system is examined through kinematic studies and structural evaluation. VGS and other digital tools are used to understand stability, deformation, and the performance of the deployable frames.

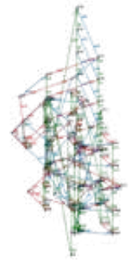
Structure Analysis

VGS

Form Diagram

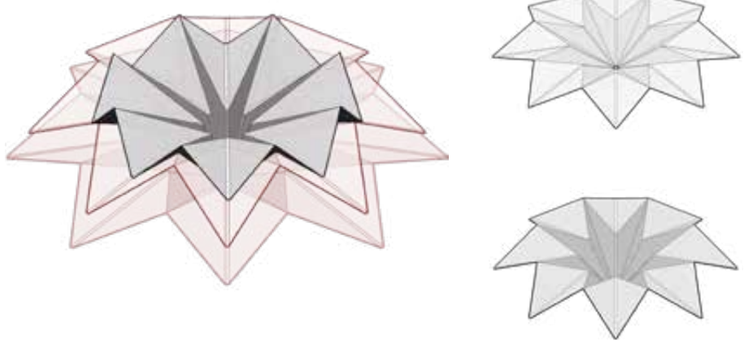


Force Diagram



Kinematic Analysis

Multiple Folding States



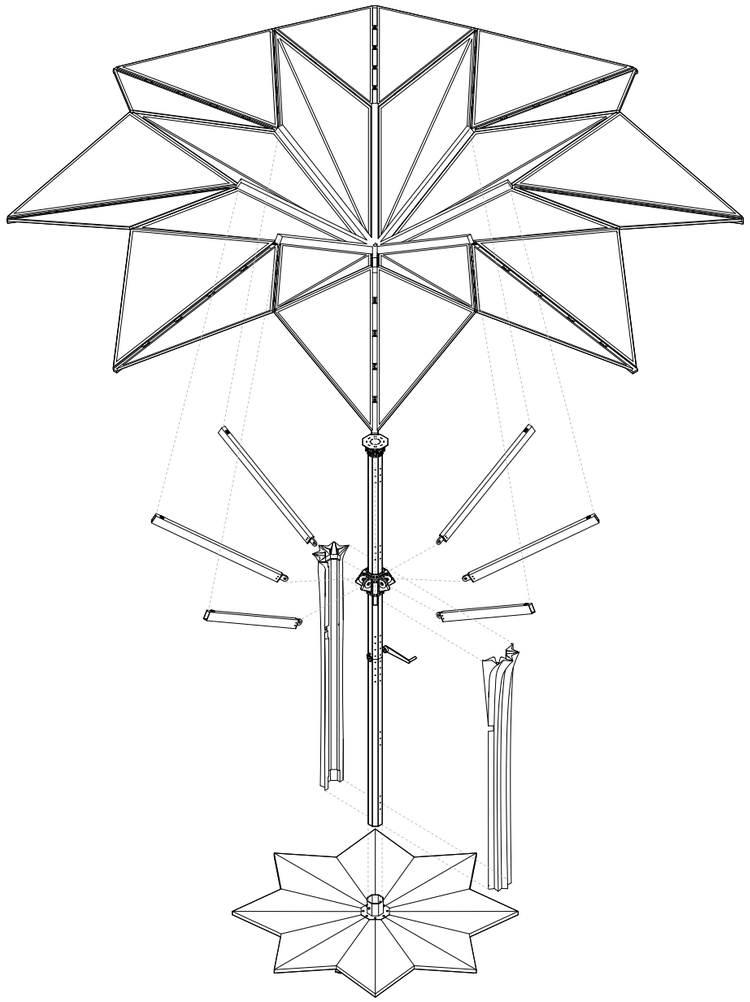
Kinematic Data

Fold Angle | Hinge Freedom | Panel Position | Self-intersection Detection

Construction

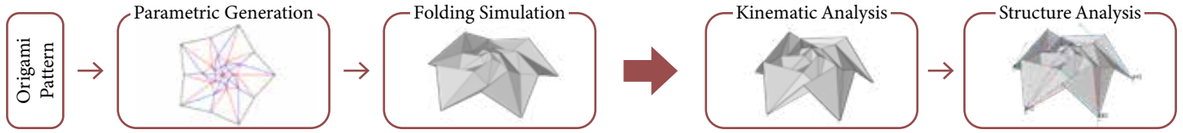
Construction details and prototyping translate the folding logic into a physical mechanism. Aluminum frames, polycarbonate panels, steel hinges, and tension ropes are assembled to realize the final deployable umbrella system.

Assembly



Origami Logic | *From Parametric Generation to Structural Analysis*

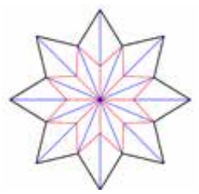
This page explores the interplay between foldable geometry and structural analysis. Parametric models and folding simulations test each shape, while Visual Graphic Statics evaluates its force flow—creating a real-time feedback loop where form and force are co-optimized.

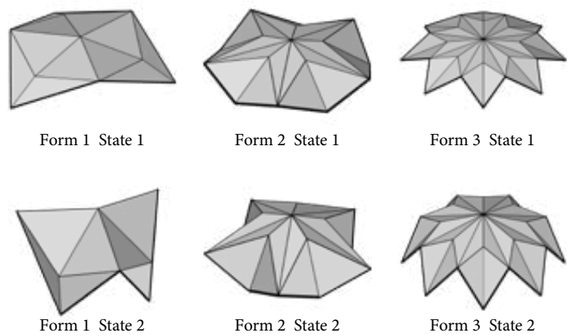


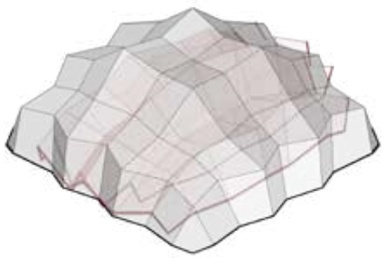
Patterns

Circular Yoshimura

Exploring planar-to-dome transformation through radial array of Yoshimura pattern.

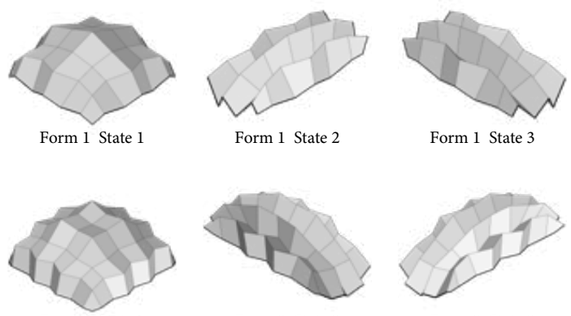


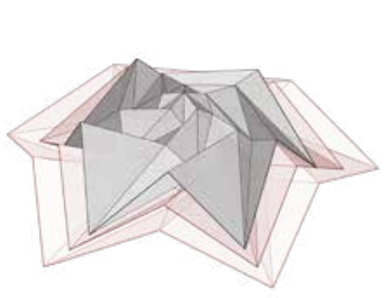




Eggbox

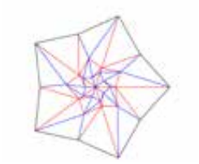
Forming developable synclastic surfaces through arrayed protruding units.

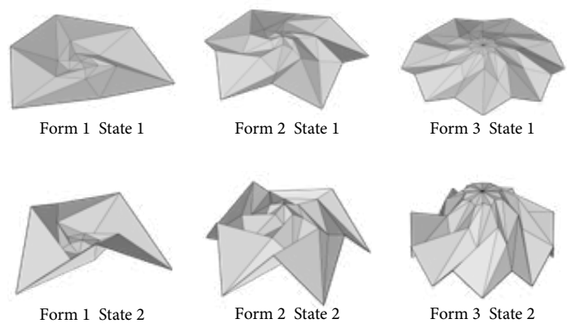


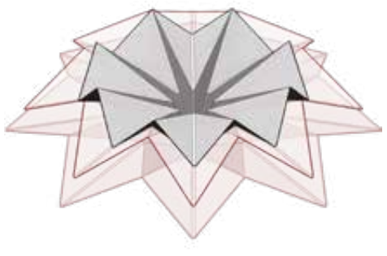


Polygonal Twist

Expanding morphological possibilities by integrating polygonal units into classical square twist.

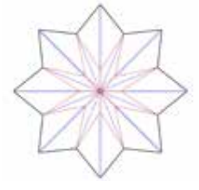


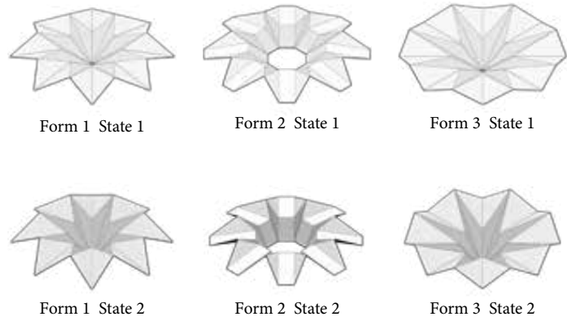




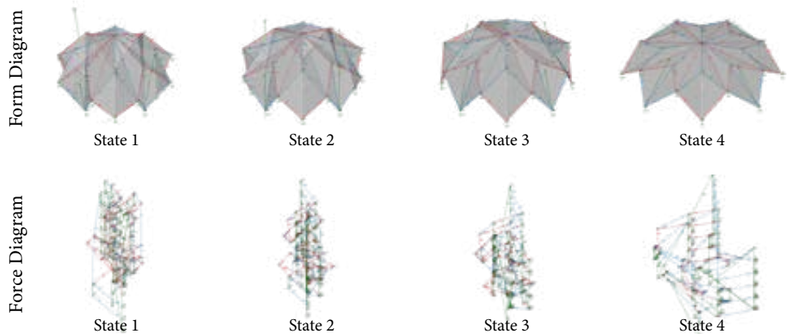
Umbrella Yoshimura

Adapting crease patterns for thick materials while achieving circular transformation through radial arrangement.





Dynamic Structure Analysis

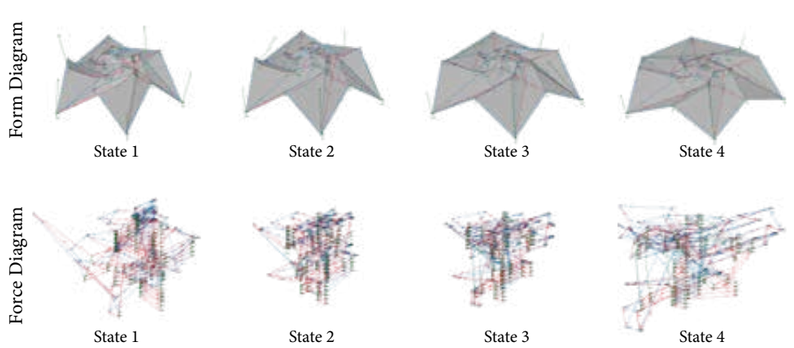


Analysis reveals significant force redistribution throughout the folding process.

EQUILIBRIUM MATRIX 75x76
S (Self Stresses) = 1
M (Internal Mechanisms) = 0
The structure is statically indeterminate
1 self-stresses have to be assigned as parameters

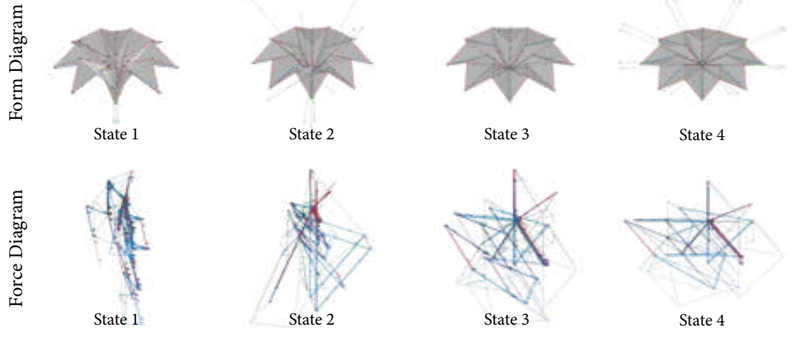
Force distribution exhibits high dynamic complexity during folding.

EQUILIBRIUM MATRIX 147x148
S (Self Stresses) = 1
M (Internal Mechanisms) = 0
The structure is statically indeterminate
1 self-stresses have to be assigned as parameters



Pivot points experience continuous directional shifts in force.

EQUILIBRIUM MATRIX 93x93
S (Self Stresses) = 0
M (Internal Mechanisms) = 0
The structure is statically determinate
0 self-stresses have to be assigned as parameters



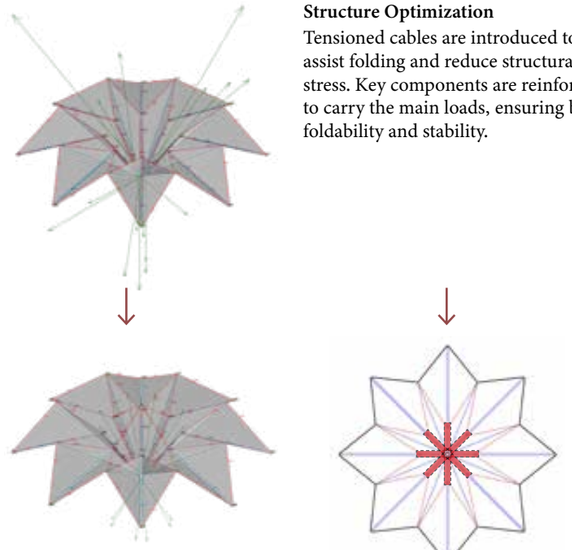
Overall stress distribution remains favorable during folding, though localized overloading occurs at specific pivot points.

EQUILIBRIUM MATRIX 192x192
S (Self Stresses) = 0
M (Internal Mechanisms) = 0
The structure is statically determinate
0 self-stresses have to be assigned as parameters

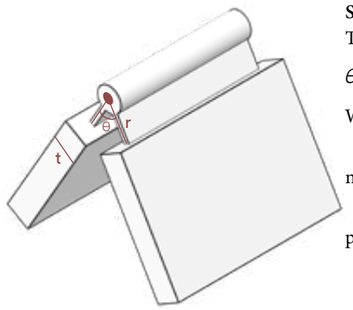
Detail Build | *Structural Insights Guide Final Assembly*

This stage presents design details and construction. Tensioned cables and reinforced members ensure foldability and load-bearing performance. Exploded views and assembly photos show how analysis guides buildability.

Optimization



Structure Optimization
Tensioned cables are introduced to assist folding and reduce structural stress. Key components are reinforced to carry the main loads, ensuring both foldability and stability.



Self-Intersection Check
The critical angle is defined as

$\theta_{crit} = 2\arctan(t/2r)$

When

- $\theta > \theta_{crit}$

no self-intersection occurs.

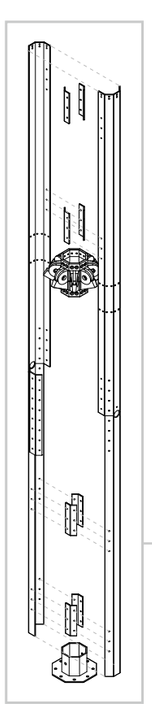
- $\theta \leq \theta_{crit}$

panels intersect.

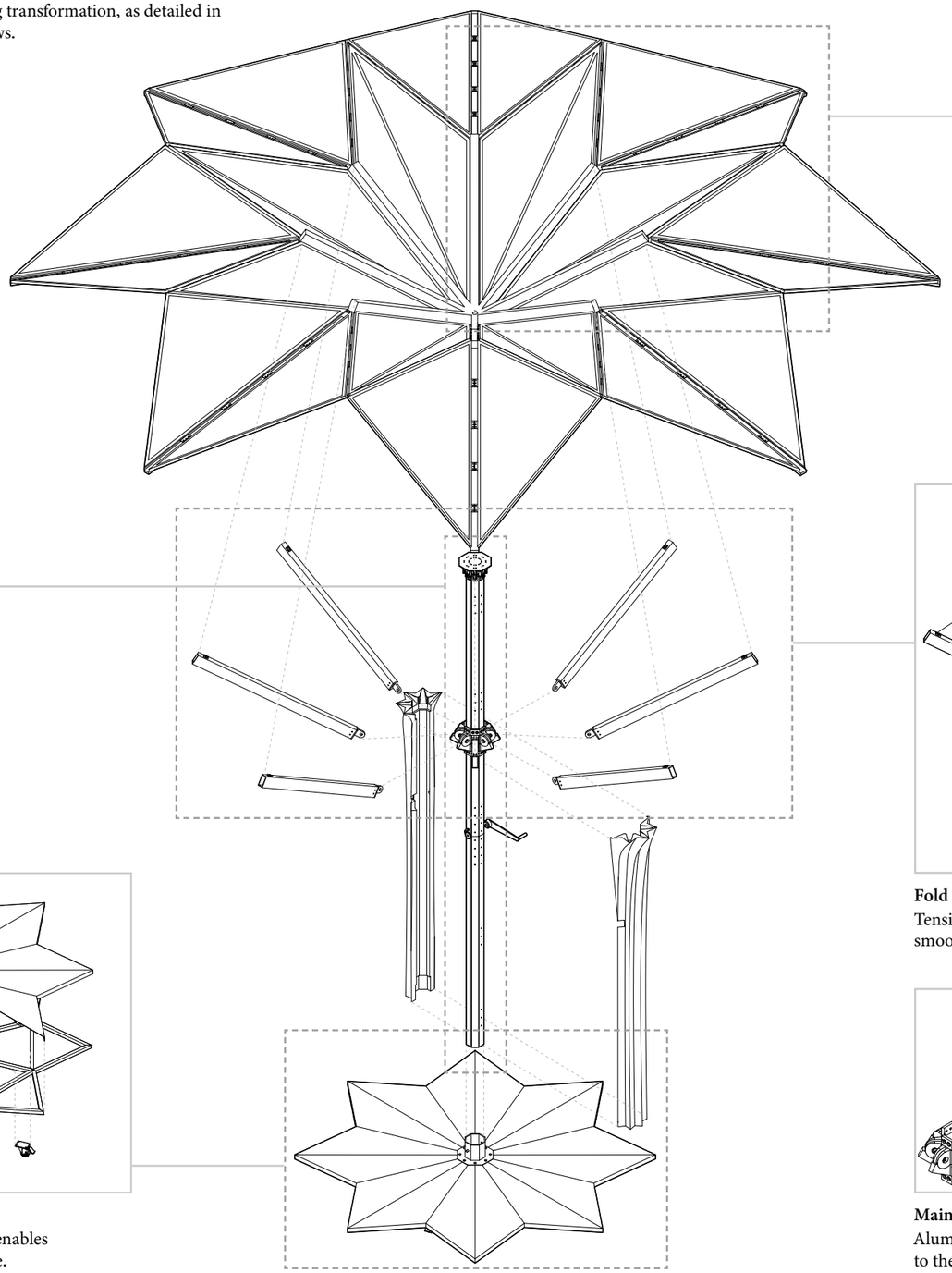
Assembly Photos

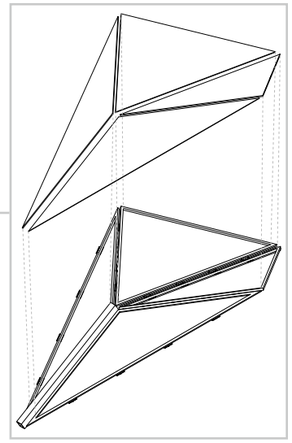
Details

This design integrates rigid panels and precision hinges for smooth deployment and stable load-bearing. Optimization of joints, structure, and materials ensures integrity during transformation, as detailed in component studies and exploded views.

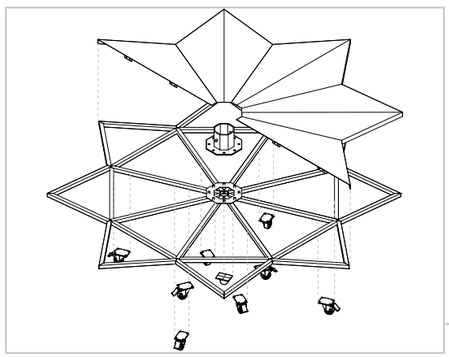


Column
Metal column providing the main structural support for the entire system.

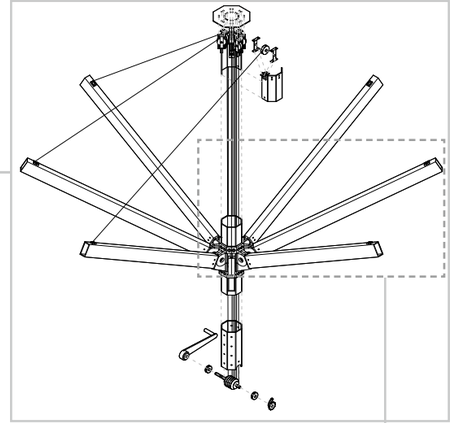




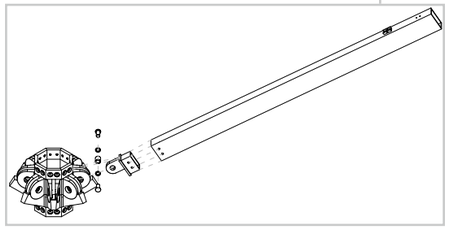
Canopy Unit
Rigid panels with aluminum frames and hinges allow easy material replacement and functional flexibility.



Base
Octagonal base with lockable wheels enables mobility and echoes the canopy shape.



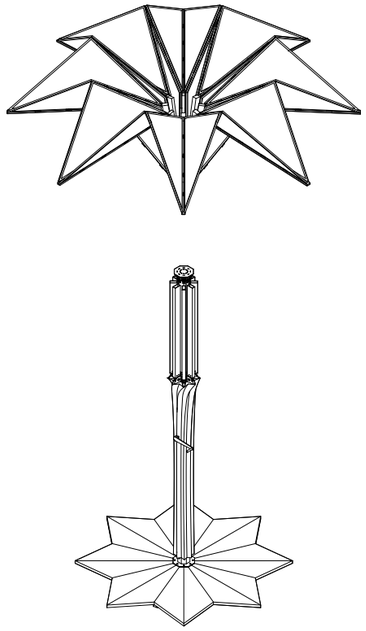
Fold Drive
Tensioned cables, guided by levers, drive smooth folding and deployment.



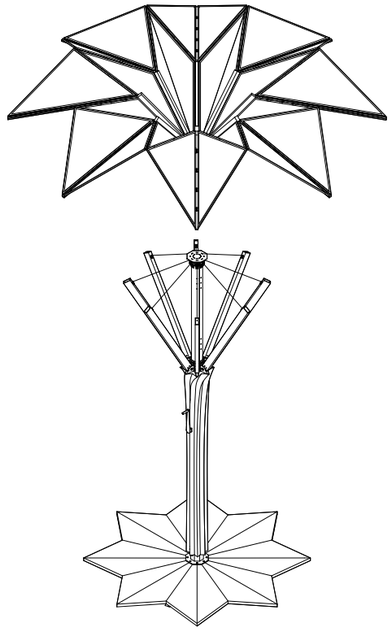
Main Component
Aluminum tube bearing main loads, connected to the column via 3D-printed ABS core.

Final Realization | *Built Structure and Folding States*

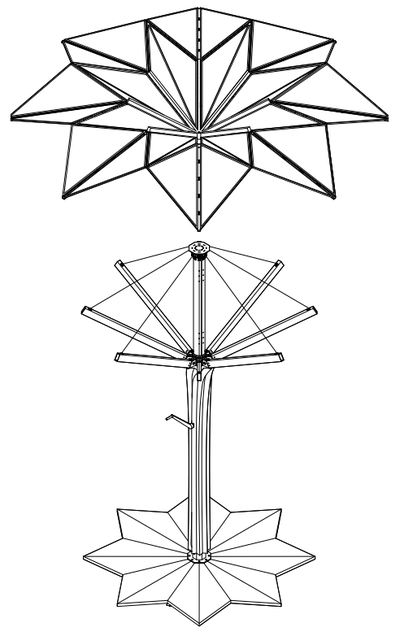
This page presents the completed structure, showing both the physical assembly and the system's folding behavior. Construction photos reveal the built form in real space, while line drawings capture three key folding configurations, illustrating the design's transformation and structural logic.



State 1



State 2



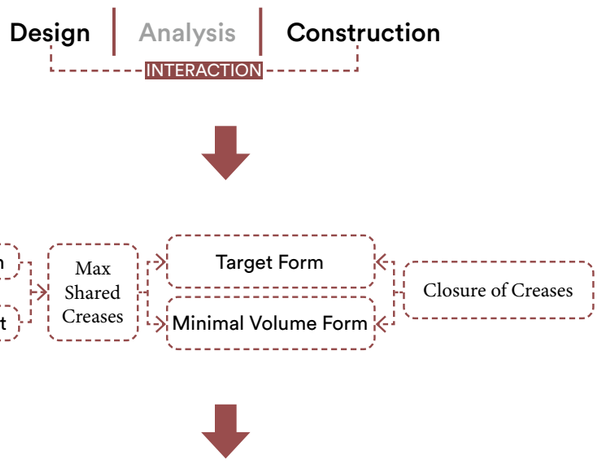
State 3

Future Research | *Integrating Cross-Stage Feedback into a Continuous Folding System*

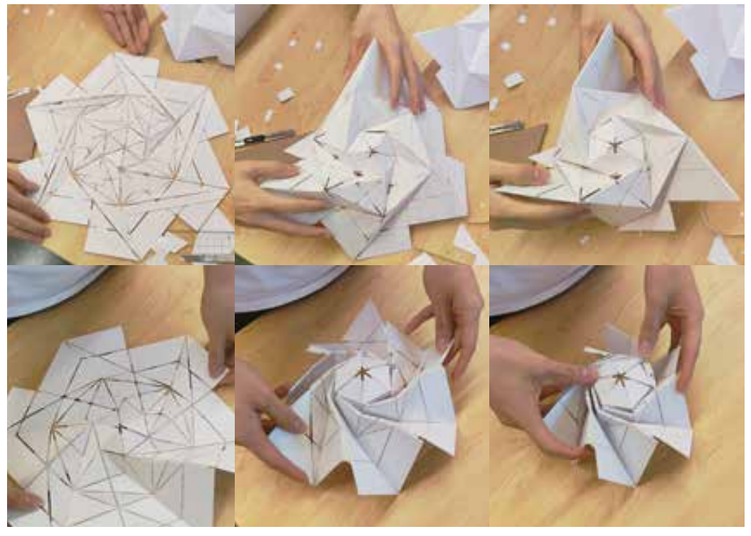
These future directions seek to dissolve the boundaries between design, analysis, and fabrication, allowing each stage to inform and reshape the others. By enabling crease systems to stack, actuate, and structurally self-evaluate, the workflow becomes a feedback-rich loop rather than a linear pipeline. Hybrid folding layers, motorized deployment, and VGS-guided optimization collectively illustrate a system where geometric intention, structural behavior, and material construction continuously interact to generate more adaptive folding outcomes.

Hybrid Crease Strategy

By combining flasher folds for compactness and polygonal-twist folds for geometric shaping, the system can switch modes through shared creases defined in design and selectively activated during construction.

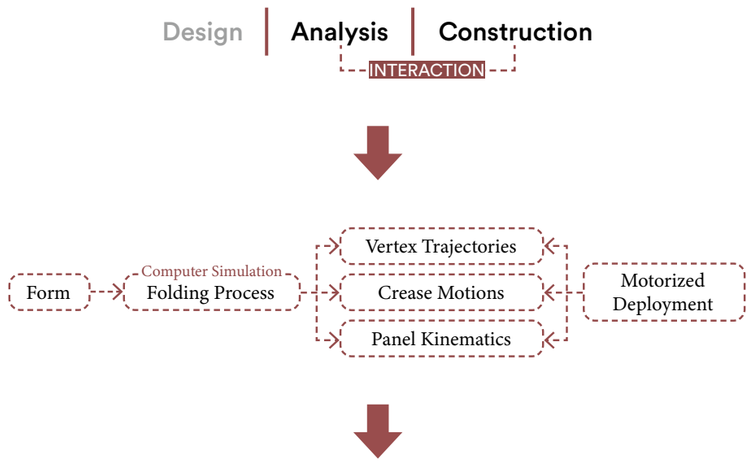


Multi-Function

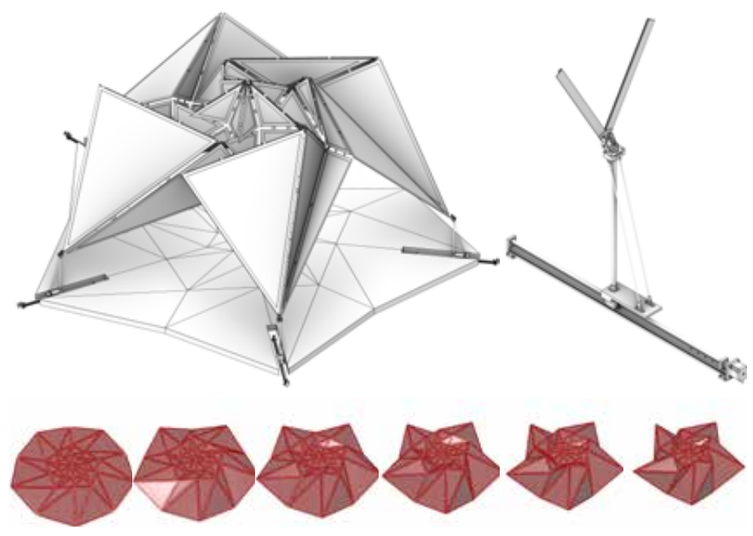


Motorized Deployment

Motor actuation enables coordinated automatic deployment, such as synchronizing the five feet of a pentagonal twist toward the center.

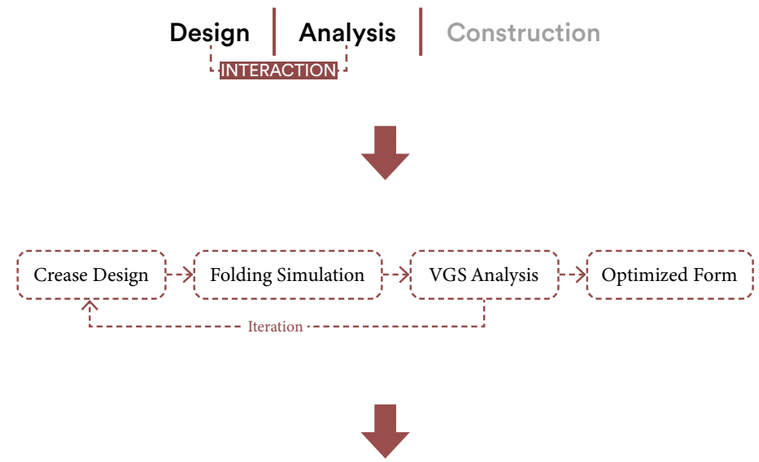


Automation

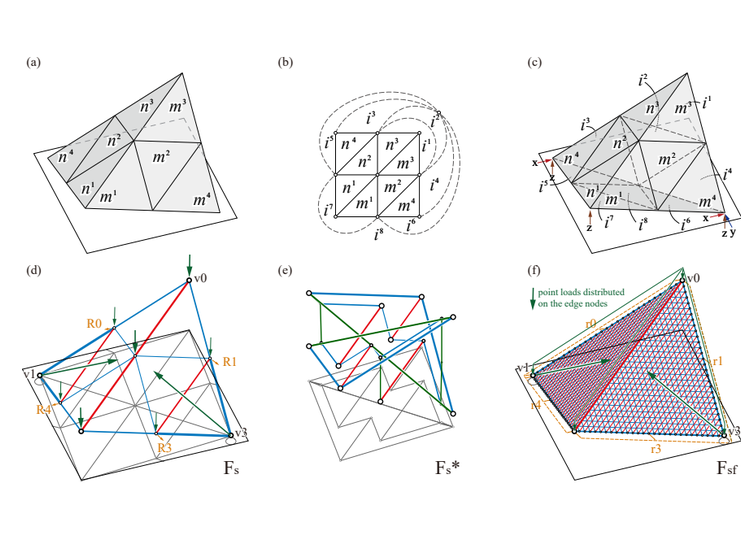


VGS-Driven Crease Optimization

Linked crease motion can be evaluated through VGS to reveal structural responses. This feedback enables iterative adjustment of the crease pattern toward a more efficient structural configuration.



Performance-Driven



Willow Path Design

Multi-Objective Optimization of Finite-Length Willow Paths

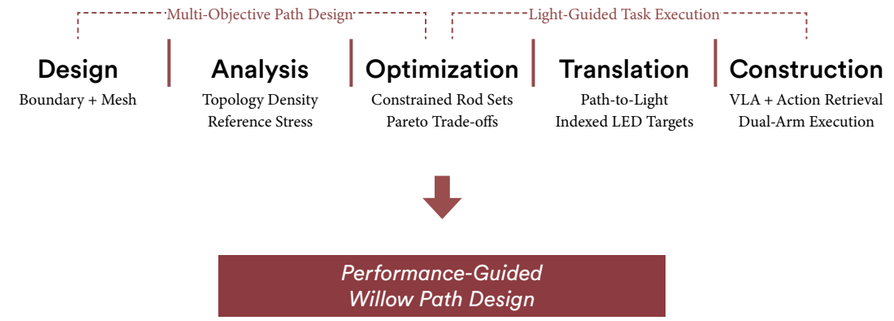
Academic Project

This project develops a computational method for designing finite-length willow paths on a fixed mesh. Topology analysis defines coverage priorities and optional rewards; reference stress directions guide path orientation.

My contribution is the constrained four-objective model and search framework: maximize optional-node benefit and stress alignment, while minimizing internal bending and rod count. Local path edits and regional reconstruction jointly adjust rod allocation and routing, preserving feasible Pareto trade-offs for design selection.

LED-based path translation and robotic weaving provide a downstream application. The current robot demonstrations use a simple pattern; execution of the optimized designs remains to be validated.

Keywords: Multi-Objective Optimization, Willow Path Design
Instructor: Yige Liu, Feng Yuan
Role: Computational Design + Multi-Rod Path Optimization
Team: Suming Li, Zhengcheng He, Guowei Zhang, Wanting Hong
Contribution: Topology / Stress Pipeline, Constraint Model, Search Algorithm, Result Visualization
Year: 2026



Workflow | *From Reference Fields to Weaving Paths*

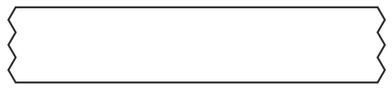
The core workflow transforms a chosen boundary and mesh into topology and reference-stress guidance, then searches constrained multi-rod path sets across four objectives. The Pareto archive supports design selection. Path-to-light translation and learned weaving actions connect these outputs to a downstream fabrication workflow.

Design

Define the planar design domain and choose how it is subdivided.

Form

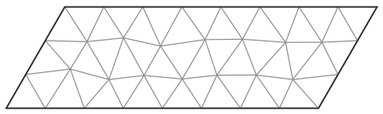
Choose a 2D outline; vary its silhouette before discretization.



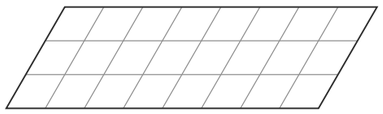
Planar boundary / no mesh

Mesh

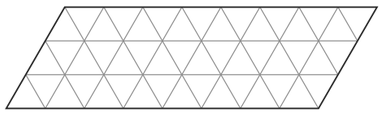
Different subdivisions of the same simple domain.



Irregular triangles / current



Quadrilaterals / alternative



Equilateral triangles / alternative

Analysis

Analyse topology and stress on a fine mesh; map guidance to the coarse weaving graph.

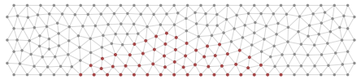
Topology

Optimize and clean the fine-grid reference; derive coarse-node rewards.

Optimize

Clean

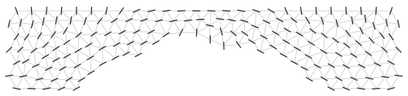
Score



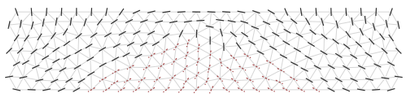
Pre-cleanup density -> coarse-node reward

Reference Stress

Analyse retained material; extend tensors and map reference axes to coarse nodes.



FE axes / retained region



Tensor extension / removed region

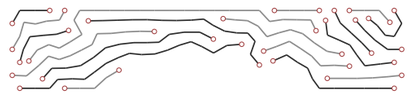
Estimated guidance; no extra FE solve.

Optimization

Search finite rod sets while preserving trade-offs between four objectives.

Multi-Objective Model

Evaluate a variable set of rods within a hard-constrained feasible space.



T Benefit +

Optional density reward

A Alignment +

Stress-axis agreement

B Bending -

Internal curvature cost

k Rod count -

Number of finite rods

Hard limits: coverage / length / curvature / path topology.

Search

Maintain feasible and repair populations; archive strictly feasible trade-offs.

Initialize + classify rod sets

↓

Local edits + repair + ALNS

↓

Strict check + Pareto archive

↓

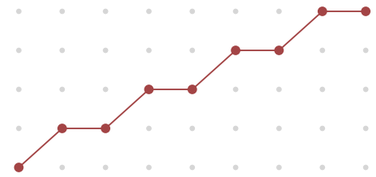
Select and export rods

Translation

Turn ordered paths into visible targets and prepare the physical assembly.

Lighting

Encode each rod's node order as indexed LED targets for robot vision.



Rod / node encoding

↓

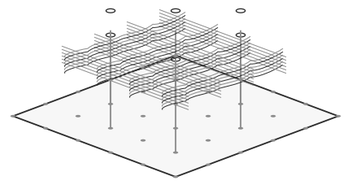
Indexed light sequence

↓

Visual target prior

Preparation

Prepare the board, guide pegs and willow assembly around a fixed spatial reference.



Board + guide pegs
Willow layers + removable caps

Construction

Learn task primitives from demonstrations; coordinate dual-arm weaving through vision, policy selection and feedback.

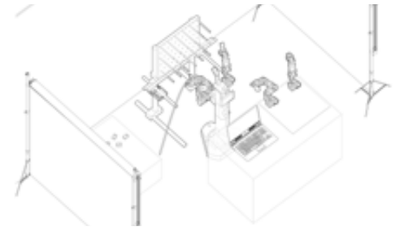
Task Learning

Collect three-view demonstrations and task instructions for four weaving primitives.

Pick	Wind upper
Wind lower	Press

Three-view images + task language
VLA training for four primitives

Robot System



LED-guided CV + tracking

↓

Scene state integration
CV + live views + task + robot state

↓

VLA reasoning | Action-library retrieval

↓

Weighted policy selection

↓

Motion command output

↓

Dual-arm weaving

Optimization | *Four-Objective Willow Path Optimization*

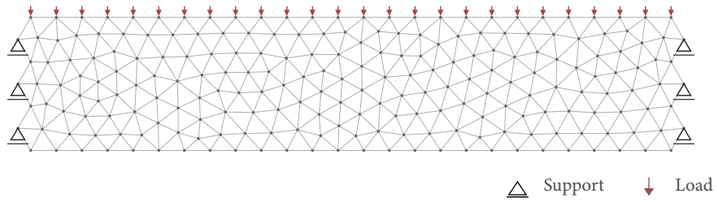
This case uses topology and reference stress to guide four-objective path optimization.

Later development separates fine-grid analysis from coarse-grid weaving and adds a feasible/repair dual-population search.

Structural Analysis | Four Reference Fields

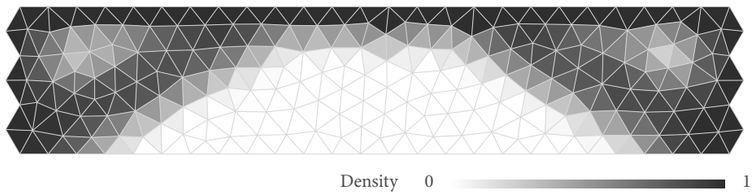
Mesh | Boundary Setup

Triangular mesh with prescribed supports and loads.



Density | Topology

Filtered and projected topology density represents material distribution.

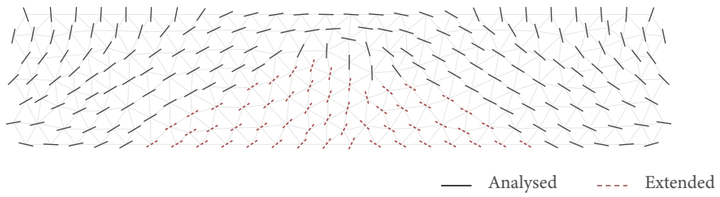


Stress | Signed Field

Reference normal stress distinguishes tension from compression.

Axes | Stress Extension

Analysed and extended stress axes provide directional guidance.

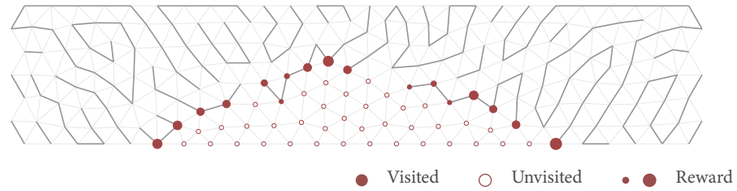


Four Objectives | Jointly Evaluated

T | Maximize Benefit

W: visited nodes; O: optional nodes.
Rewards counted once / fixed total.

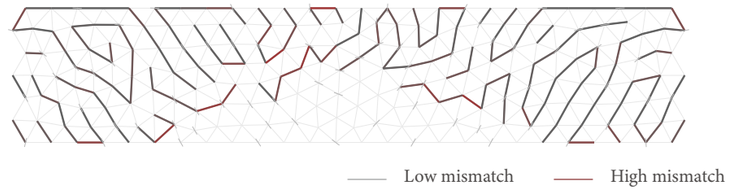
$$T(\mathcal{R}) = \frac{\Sigma_{v \in W \cap O}\, r_v}{\Sigma_{v \in O}\, r_v}$$



A | Maximize Alignment

E: used edges; w: stress magnitude.
d: directional mismatch.

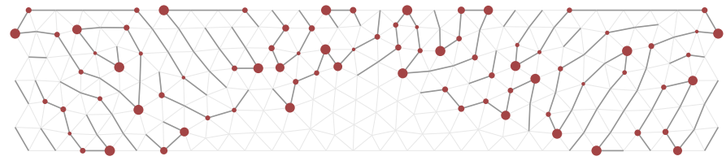
$$A(\mathcal{R}) = 1 - \frac{\Sigma_{e \in E}\, w_e\, d_e}{\Sigma_{e \in F}\, w_e}$$



B | Minimize Bending

Geometric proxy; joints excluded.
Squared turn / mean adjacent length.

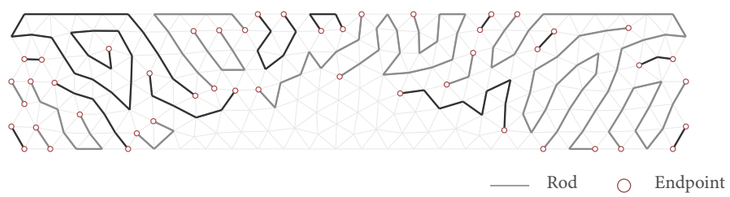
$$B(\mathcal{R}) = \Sigma_{j=1}^{k}\, \Sigma_{i=1}^{n_j - 2}\, \frac{\theta_{j,i}^2}{\bar{\ell}_{j,i}}$$



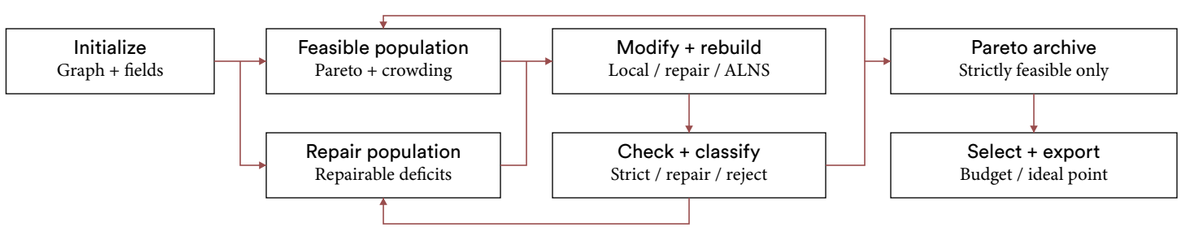
k | Minimize Rod Count

Number of rods; limits fragmentation.

$$k(\mathcal{R}) = |\mathcal{R}|$$



Optimization Framework | Multi-Path Pareto Search



Four-Objective Trade-offs | Pareto Dominance

$$\mathbf{f}(\mathcal{R}) = (-T(\mathcal{R}), -A(\mathcal{R}), B(\mathcal{R}), k(\mathcal{R})), \quad \mathcal{R} \in \Omega.$$

Dominance: no worse in all four raw objectives, strictly better in at least one; feasible sets only.

Retain evaluated non-dominated candidates. Weights select display plans only.

Hard Constraints | Illustrated Case

Graph + Coverage

164 mandatory nodes; ≤20 unique optional nodes.
Simple open paths: no reused edges, branches or cycles.
≥1 edge / rod; shared nodes join two rod ends only.
End / joint permissions apply; prescribed ends stay exposed.

Material + Geometry

Per rod: Lgeo ≥30; Lmat = Lgeo +40 ≤1500.
20 reserved / end; internal curvature ≤1.
Turns: internal ≤100°.
Lengths: model units; curvature: inverse length.

Mutation Operators | Six Local Moves

Split
Divide one rod into two.

Merge
Join adjacent rod ends.

Relocate
Transfer a node between rods.

Swap tails
Exchange tails between rods.

Add / remove
Insert / remove optional nodes.

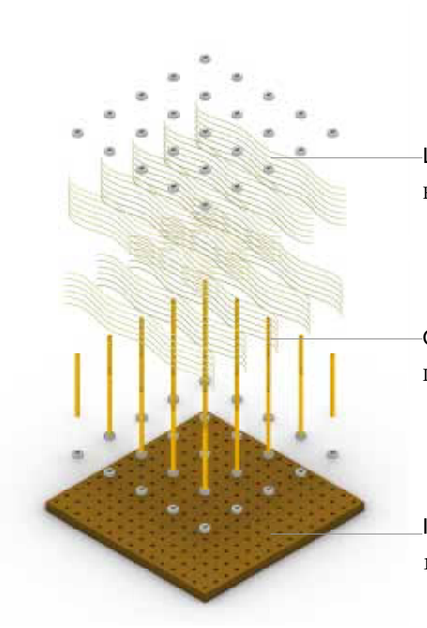
Reorder
Reverse a local subpath.

System Integration | *Path Translation + Weaving Setup*

Path-to-light translation connects the computational output to an indexed weaving workspace. The panel, guides and LED targets establish a shared reference. A collaborative robot system combines visual cues, learned task primitives and retrieved actions for dual-arm weaving. It is a downstream fabrication study; the optimized path designs have not yet been executed by the robot.

Willow Panel

Layered rods are located by removable guides and secured to an indexed board. Caps fix the assembly and allow release.

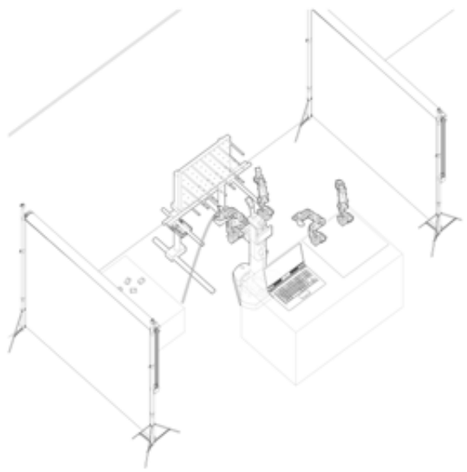


Layered Willow

Ordered rods form successive layers. Guides locate the paths; caps fix the assembly.

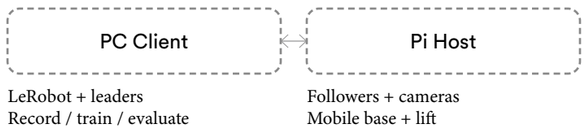
Experimental Setup

Dual AM-ARM200 Pro follower arms, a mobile base and a motorized lift. LeRobot provides the software interface.

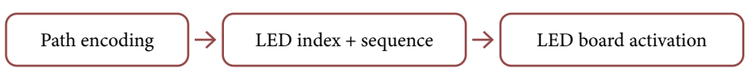


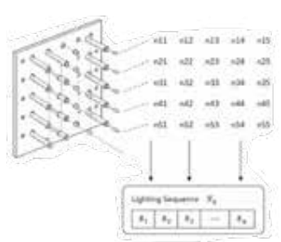
Experimental workspace / robot + cameras

LeRobot Interface

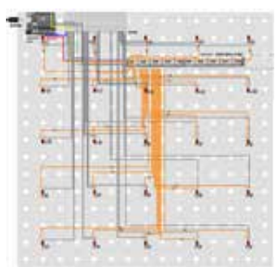


Path-to-Light Control



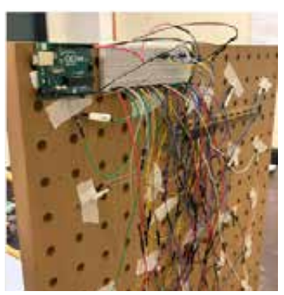


Lighting Sequence



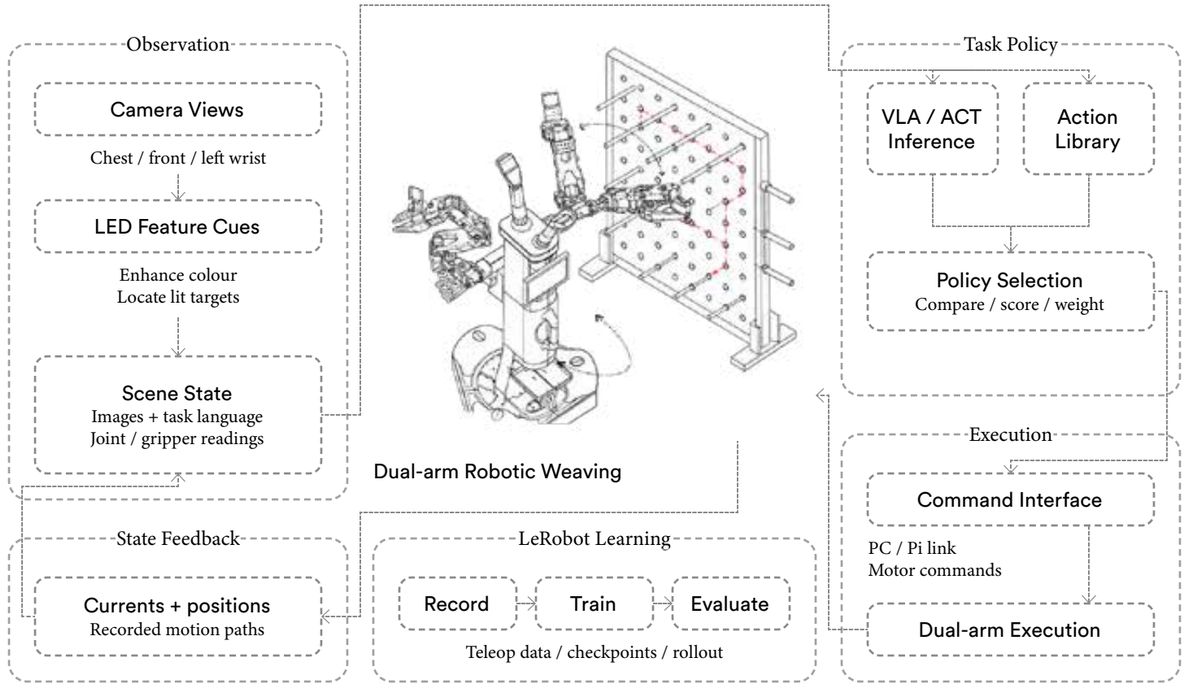
UNO R3 + LED drivers

Control System

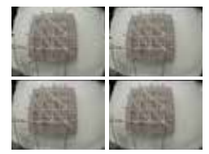
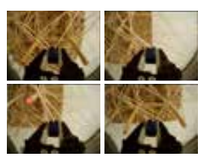


LED board prototype

Robot System Framework



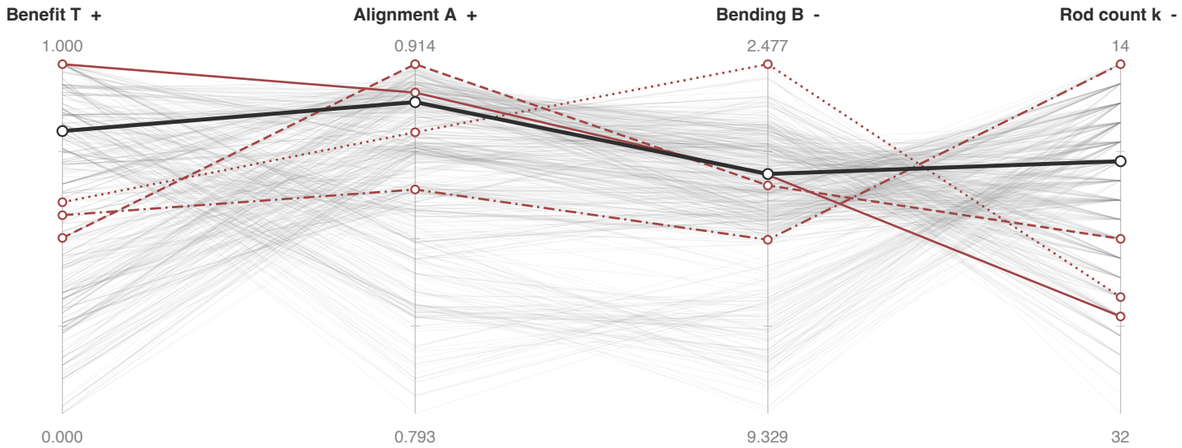
Task Primitives + Training Records

Action Demonstration	Episodes / frames per ep.	Chest View	Front View	Left Wrist
	Material Picking Grasp the blue block. 50 / 500-800			
	Upper Winding Wind above red targets. Move left to right. 200 / 1000-1600			
	Lower Winding Wind below red targets. Move left to right. 200 / 1000-1600			
	Pressing Push into the gap between two red targets. 50 / 900-1200			

Results | *Pareto Trade-offs + Morphology Studies*

The saved Pareto archive compares feasible willow path sets across four objectives. Representative designs reveal their trade-offs; six morphology studies explore different outlines and mesh types. Robot frames show four weaving primitives on a simple pattern, providing a separate fabrication study rather than validation of the optimized designs.

Pareto Archive



Every axis: better toward the top. All 599 candidates shown.

T max — 01 / archive #001

T 1.000 / A 0.904
B 4.651 / k 27

A max — 02 / archive #371

T 0.503 / A 0.914
B 4.856 / k 23

B min — 03 / archive #344

T 0.605 / A 0.890
B 2.477 / k 26

k min — 04 / archive #356

T 0.568 / A 0.870
B 5.921 / k 14

Weighted sum — 05 / archive #206

T 0.809 / A 0.901
B 4.631 / k 19

Front-Based Display Selection

Normalize within the final archive; select the smallest squared distance to its ideal point. Search and dominance remain unchanged.

$$X^* = \arg\min_{X \in A_{\text{final}}} \sum_{i=1}^{4} z_i(X)^2$$

$$z_i(X) = [\, f_i(X) - f_i^{\min} \,] \,/\, [\, f_i^{\max} - f_i^{\min} \,]$$

Equal weights; zero-range loss = 0.

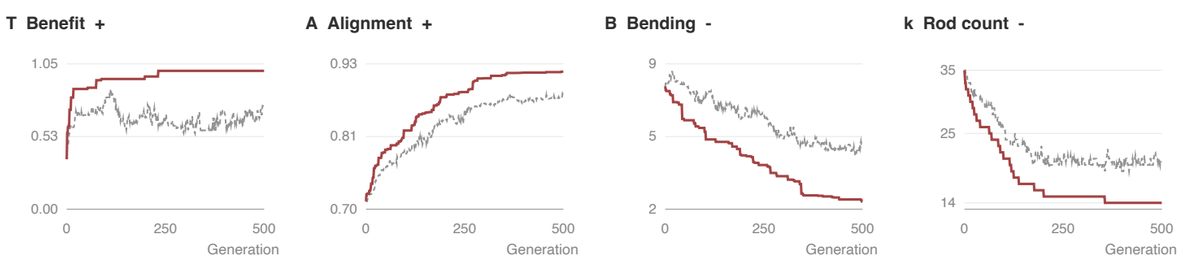
Search Progress

599
Non-dominated solutions

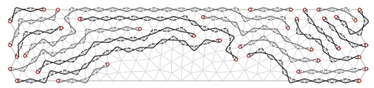
31,839
Unique feasible plans evaluated

500
Completed generations

98.6°
Max internal turn (all plans)

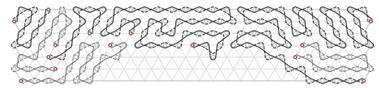


Morphology Studies



Irregular Triangles

T 0.809 / A 0.901 / B 4.631 / k 19

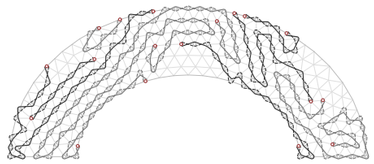


Equilateral Triangles

T 0.838 / A 0.894 / B 16.778 / k 7

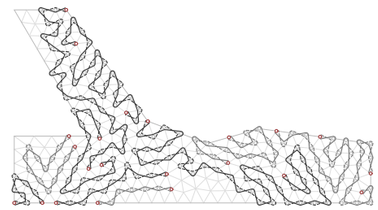
Quadrilateral Grid

T 0.658 / A 0.928 / B 29.197 / k 9



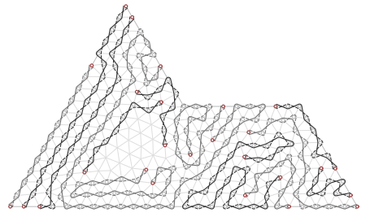
Arched Bench

T 0.828 / A 0.932 / B 15.449 / k 9



Chaise Longue

T 0.777 / A 0.837 / B 29.262 / k 11



Chair

T 0.423 / A 0.869 / B 20.358 / k 13

Robotic Weaving

Grasp

The robot grasps and lifts the willow.

Upper wrap

The robot winds the willow above the target.

Lower wrap

The robot winds the willow below the target.

Push

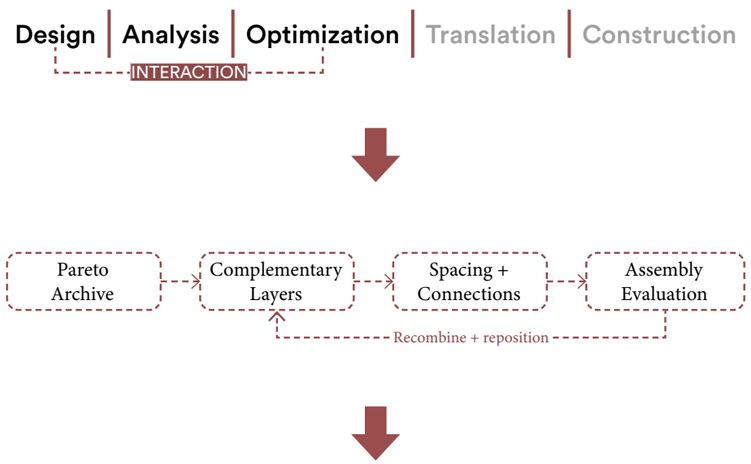
The robot presses the willow into the gap.

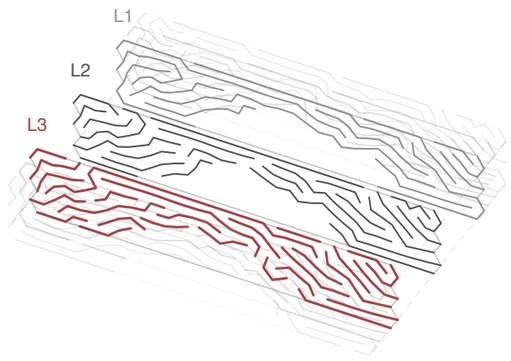
Future Research | *From Geometric Paths to Validated Weaving*

The discussion identifies three directions for advancing the current single-layer path model: layered assembly design, material and support calibration, and robotic process validation. Each requires physical evidence alongside geometric evaluation. The present archive describes trade-offs among evaluated candidates; it does not establish global optimality or structural performance.

Layered Assembly

Use the single-layer Pareto archive as a library for layered design. Select complementary paths, then evaluate their combined coverage, alignment and rod allocation with layer thickness, connections and interference.

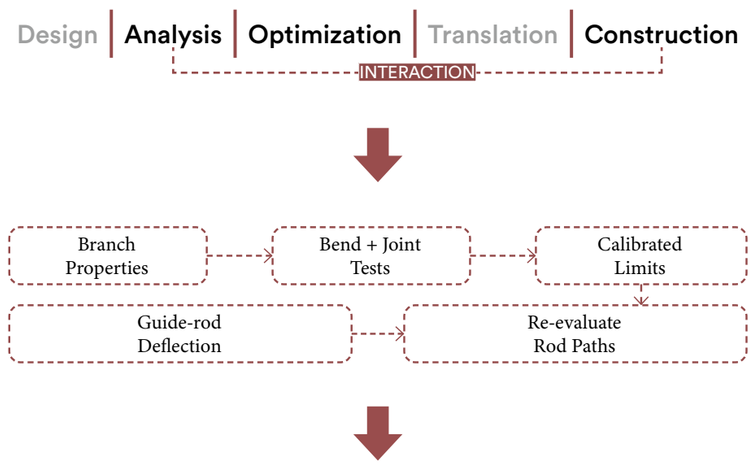


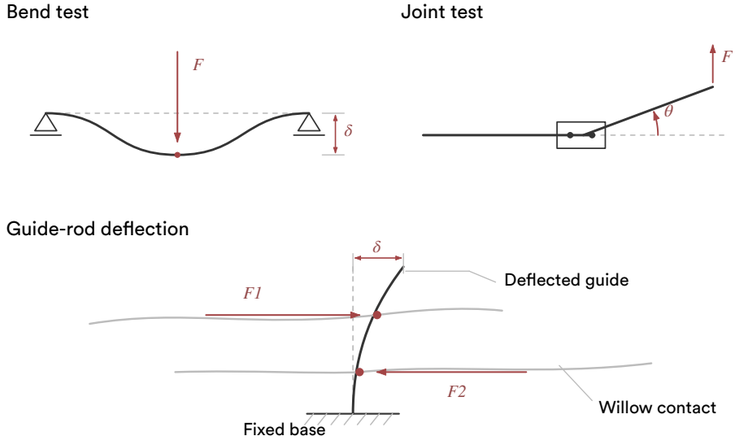


L1 Rod count / L2 Alignment / L3 Benefit

Material Calibration

Replace uniform branch limits with measurements of effective length, diameter, moisture and defects. Calibrate bending and joint behaviour, and measure guide-rod deflection under staged contact loads.

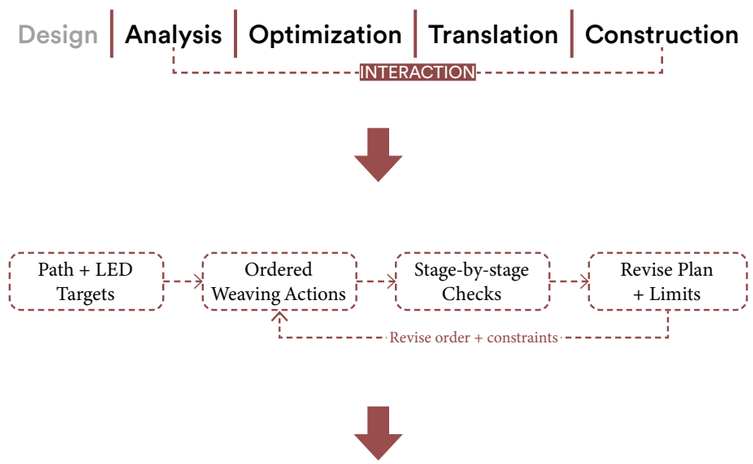


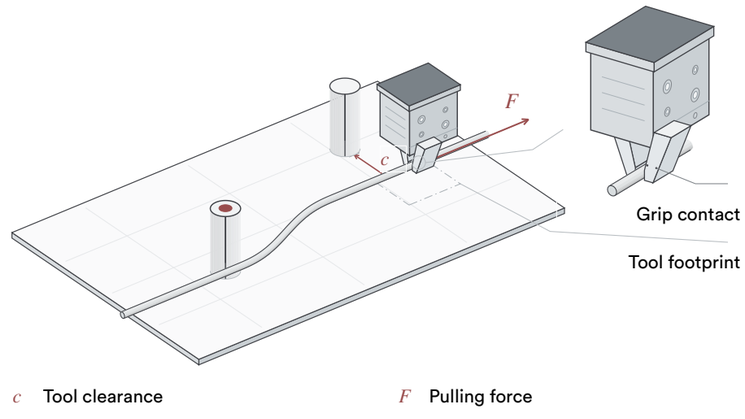


Proposed test setups

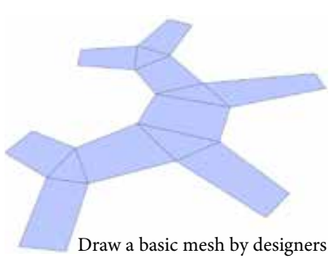
Robotic Process

Convert geometric paths into tested operations. Coordinate tool clearance, intermediate willow shape and pulling demand with grasp capacity and weaving order; final-path feasibility alone cannot establish process feasibility.

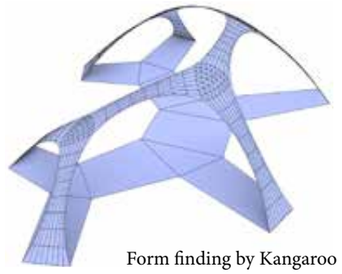




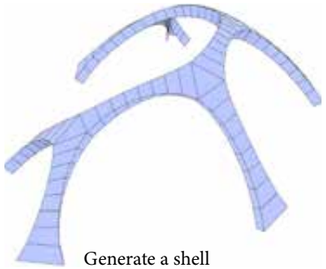
Proposed process study / tool reference: Bandy2 (IAAC)



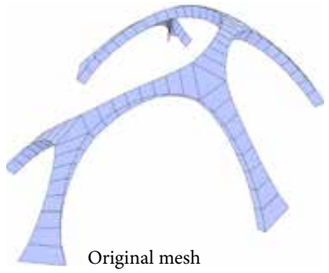
Draw a basic mesh by designers



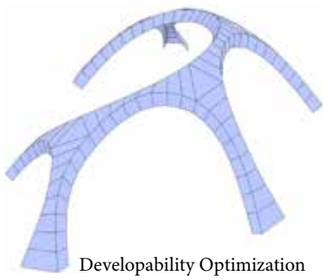
Form finding by Kangaroo



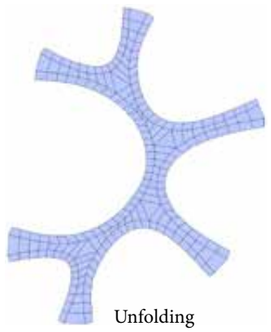
Generate a shell



Original mesh



Developability Optimization



Unfolding

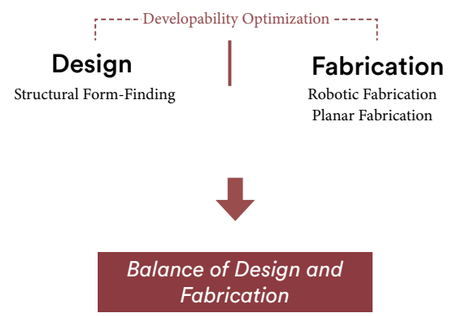
Curve Folding

Curve-Crease Shell Fabrication using Developable Approximation

Professional Project

This project investigates a computational approach for transforming freeform shell geometries into developable and foldable curved folding structures while maintaining geometric continuity and design intent. It establishes an integrated computational framework that connects shell form generation, developability optimization, and fold-aware geometric control within a single workflow. The pipeline combines Kangaroo-based form finding, multi-objective mesh optimization targeting developability and shape preservation, and unfolding validation for fabrication. The project results in a reproducible design-to-fabrication pipeline and demonstrates how developability-driven geometric optimization can bridge complex shell design with material-feasible construction logic. I hope to explore workflow-driven approaches in which shell design, geometric optimization, and construction are treated as an interconnected and mutually influential process.

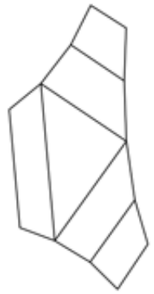
Keywords: Curved-Crease Shell, Developable Surfaces, Geometric Optimization, Robotic Fabrication
Instructor: Xiang Wang
Role: Workflow Setup + Computational Design
Team: Suming Li, Chao Yuan
Contribution: Parametric Modeling & Form-Finding, Workflow Setup, Final Output Preparation
Year: 2025



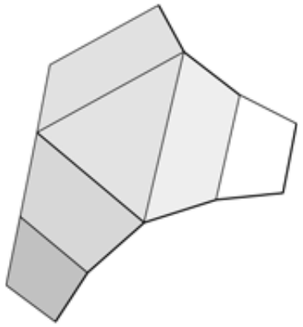
System Summary | *Breaking Down the Computational Pipeline*

This page provides a structured technical overview of the developable-shell system. It explains how the workflow, mathematical formulation, and energy components interact to guide the geometry toward a fabricable state. Each section clarifies the computational role it plays in shaping the final outcome. Together, they map the full logic of generation, optimization, and validation.

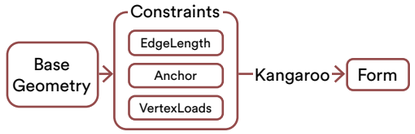
Pipeline Overview



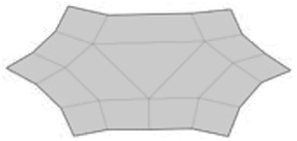
Base Geometry Setup
Starting from a simple wireframe or patch, a clean base topology is established to support reliable form derivation in later steps.



Kangaroo Form-Finding
Using Kangaroo's physics-based relaxation, the geometry evolves under simulated forces to converge toward a structurally coherent shell form.

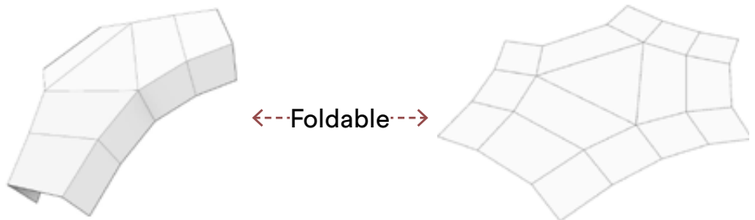


Sidewall Generation
Sidewalls are generated from the shell boundary to form a continuous and constructible spatial unit.



Unfolding
The optimized shell and sidewalls are unfolded into 2D layouts, producing fabrication-ready patterns with consistent edge lengths and fold logic.

Developable Optimization



Objective Function:

$$\min_{V} \; E_{dev} + \lambda E_{shape}$$

$$\min_{V} \; E_{\text{Gauss}} + E_{\text{planar}} + \lambda_s E_{\text{smooth}} + \lambda_i E_{\text{iso}} + \lambda_f E_{\text{fold}}$$

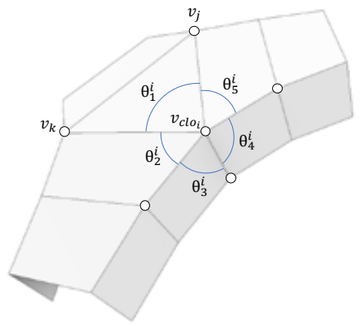
Developability Energy:
Measure shape developability

E_{Gauss} : Measure Gaussian curvature of each vertex
E_{planar} : Measure the planarity of each quad face

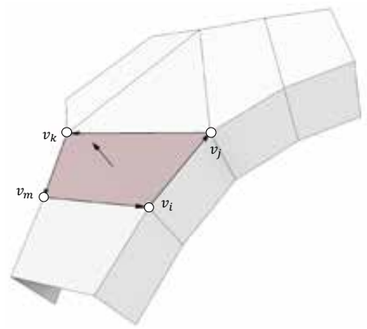
Shape Energy:
Measure shape changes

E_{smooth} : Measure the smoothness of the sidewall
E_{iso} : Measure the shape distortion
E_{fold} : Measure the change of the fold angle

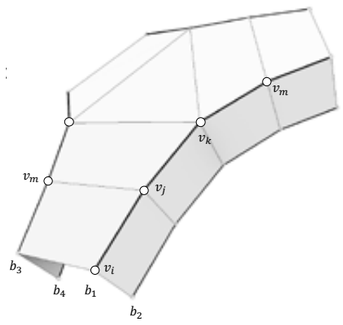
Optimization Energy Definitions



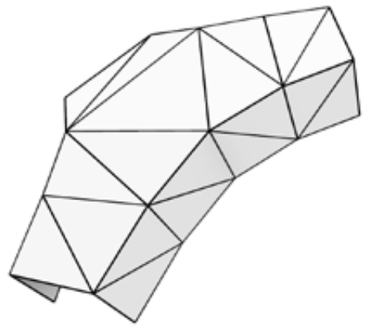
Gaussian Curvature Energy
Quantifies Gaussian curvature at each closed vertex by measuring the deviation of adjacent angle-sums from 2π; a primary term that drives the surface toward developability.



Quad Planarity Energy
Assesses quad planarity via consistency of face normals, constraining panels to be planar for fabrication-friendly geometry.

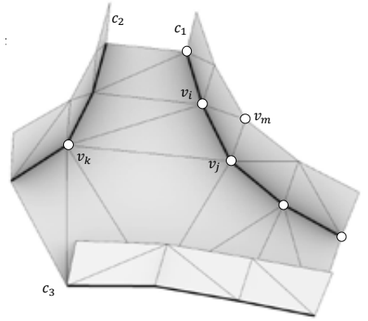


Sidewall Smoothness Energy
Compares pre- and post-optimization edge lengths as a regularizer against stretching/compression, preserving length consistency for accurate unfolding.

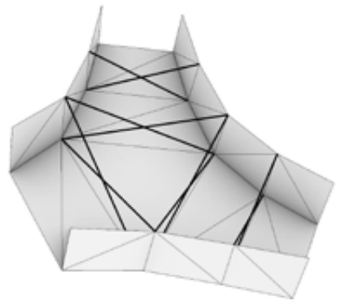


Isometric Distortion Energy

Compares pre- and post-optimization edge lengths as a regularizer against stretching/compression, preserving length consistency for accurate unfolding.

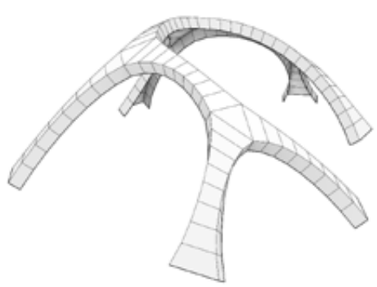


Fold-Angle Energy
Definition & Local Measure
Presents the core formulation of fold energy, defining the local measures—triangle normals and vertical edge vectors—used to evaluate changes in fold angles.



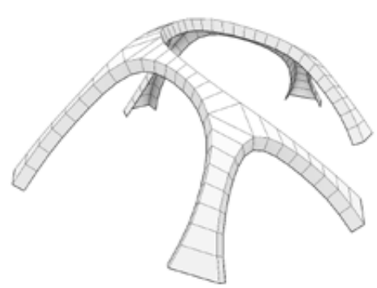
Fold-Angle Energy
Normalization & Aggregation
Describes normalization and aggregation for fold energy (diagonal/edge normalization), and explains how these terms are weighted and assembled into the total energy.

Shell Optimization Results



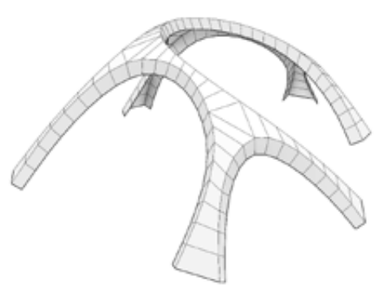
Initial Gaussian curvature error: 1.1416493330469704
Initial planarity error: 0.6476621193921162

$\lambda_s = 0.1,\quad \lambda_i = 0.1,\quad \lambda_f = 0.1$



------optimization-----
time: 59.03680992126465 iteration: 70
Gaussian curvature error: 0.0630266540290316
Planarity error: 0.032527456835975646

$\lambda_s = 0.1,\quad \lambda_i = 0.01,\quad \lambda_f = 0.01$

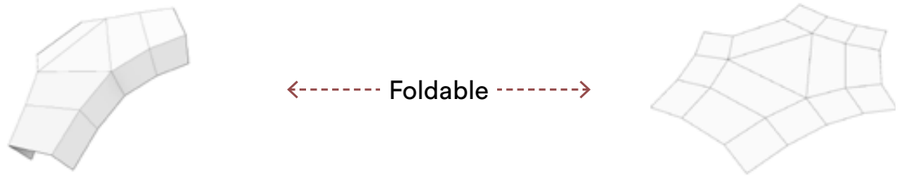


------optimization-----
time: 51.35724210739136 iteration: 62
Gaussian curvature error: 0.021614705753506266
Planarity error: 0.010024697873560856

Developability Optimization | *From Optimization to Material Exploration*

This study balances a shell's developability with the preservation of its intended form. Five energy terms combine developability and shape control, while two weight settings reveal their effect on curvature and planarity errors. Alongside this numerical study, 4D-printing experiments extend the investigation into physical form.

Optimization Objective



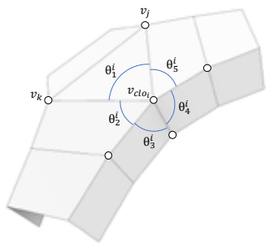
Objective Function:

$$\min_V \; E_{dev} + \lambda E_{shape}$$

$$\min_V \; E_{\text{Gauss}} + E_{\text{planar}} + \lambda_s E_{\text{smooth}} + \lambda_i E_{\text{iso}} + \lambda_f E_{\text{fold}}$$

Energy Definitions

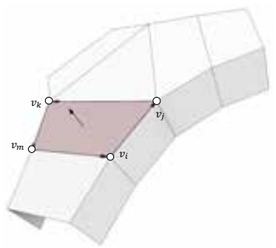
Gaussian Curvature



Interior angle sum approaches 2ϖ

$$E_{\text{Gauss}} = \sum_{i \in V_{clo}} \left(2\pi - \sum_{a_j \in A_i} \theta_j^i\right)^2$$

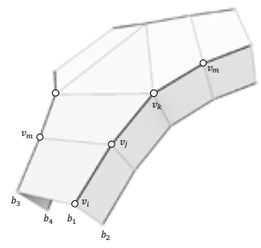
Quad Planarity



Planar quadrilateral faces

$$E_{\text{planar}} = \sum_{f_q \in F_q} \sum_{i \in f_q} \left(\bar{\boldsymbol{n}}_{f_q}^T \boldsymbol{n}_i\right)^2$$

Sidewall Smoothness



Consistent boundary bending

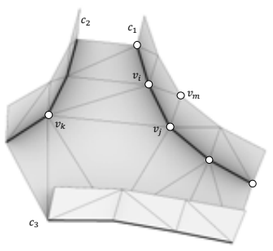
$$E_{\text{smooth}} = \omega_s \sum_{b \in B} \sum_{(j,k) \in b} ||u_j - u_k||_2^2$$

Isometric Distortion

Preserve reference edge lengths

$$E_{\text{iso}} = \frac{1}{E^2} \sum_{e \in E} \left(||v_i - v_j||_2 - ||\tilde{v}_i - \tilde{v}_j||_2\right)^2$$

Fold Angle / Hinge



Preserve crease hinge angles

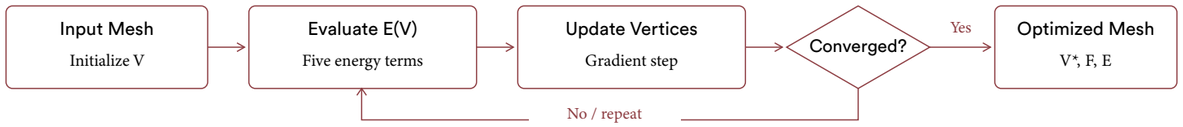
$$E_{\text{fold}} = \sum_{c \in C} \sum_{h \in c} \left(\frac{e_{ijk}^T e_{ijm}}{||e_{ijk}||_2 \, ||e_{ijm}||_2} - \cos(\tilde{\alpha}_h)\right)^2$$

Fold / Diagonal Alternative

Alternative fold term using diagonal lengths

$$E_{\text{fold}} = \frac{1}{D^2} \sum_{d \in D} \left(||v_i - v_j||_2 - ||\tilde{v}_i - \tilde{v}_j||_2\right)^2$$

Iterative Optimization

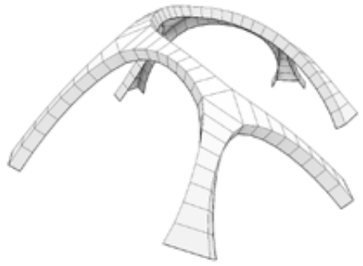


Optimization Results

Changing the shape-control weights gives different developability residuals for the same starting shell.

Original Mesh

Input geometry

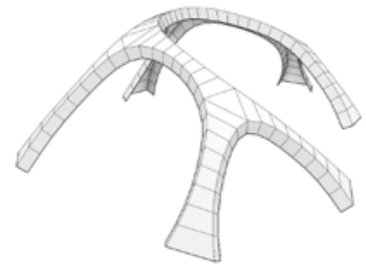


Gaussian error	1.14165
Planarity error	0.64766

Before optimization

Configuration A

$\lambda_s = 0.1 \quad \lambda_i = 0.1 \quad \lambda_f = 0.1$

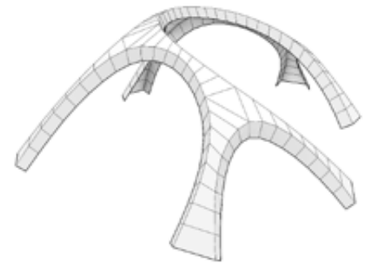


Gaussian error	0.06303
Planarity error	0.03253

70 iterations / 59.0 s

Configuration B

$\lambda_s = 0.1 \quad \lambda_i = 0.01 \quad \lambda_f = 0.01$



Gaussian error	0.02161
Planarity error	0.01002

62 iterations / 51.4 s

Configuration B relative to the original mesh

98.1%

Lower Gaussian curvature error

98.5%

Lower quad planarity error

4D Printing | Material Experiments

4D-printed samples from the material experiments.

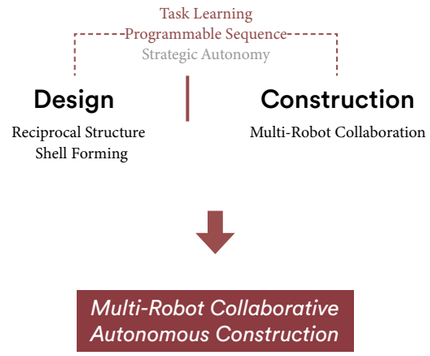
Co-Constructed Shell

Multi-Robot Collaborative Autonomous Assembly of Reciprocal Shells

Academic Project

This project investigates how a multi-robot system can autonomously learn and execute the assembly logic of reciprocal frame structures. We developed an integrated computational framework combining parametric geometric generation, programmable action decomposition, machine-learning–based action classification, and coordinated multi-robot control. The workflow incorporates Kangaroo-based form-finding, supervised multi-view action labeling, and SLAM-guided robotic execution. The resulting system demonstrates a closed-loop robotic construction process capable of interpreting and performing the full sequence of reciprocal shell assembly. This work advances a scalable model for autonomous fabrication, showing how intelligent multi-robot collaboration can bridge digital design with physical construction in architectural robotics. I am motivated to create adaptive, feedback-enabled workflows that unify computational design with robust autonomous fabrication.

Keywords: Computational Design, Robotics, Task Learning
Instructor: Xiang Wang, Zhen Xu, Ye Zhang
Role: Computational Design + ML + Robotic Development
Team: Suming Li, Xu Liu, Zihan Xue, Yu Cui, Zimeng
My Contribution: Parametric Modeling, Programmable Sequence Research, Data Collection, Details Design
Year: 2024



Workflow | *From Computational Generation to Automated Robotic Construction*

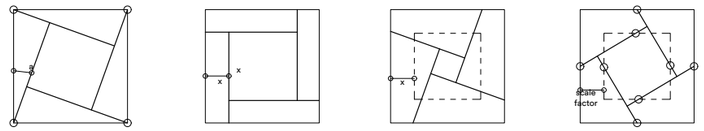
This page visually articulates a comprehensive technical workflow for reciprocal structures, spanning from digital computation to physical robotic fabrication. The process initiates with parametric algorithms for morphological generation and structural form-finding, which is then discretized into executable robotic assembly sequences. It integrates coordinated motion planning between a mobile platform and robotic arms, and culminates in a machine learning pipeline that recognizes and optimizes elemental actions—establishing a closed-loop, intelligent construction system from model to manifestation.

Design

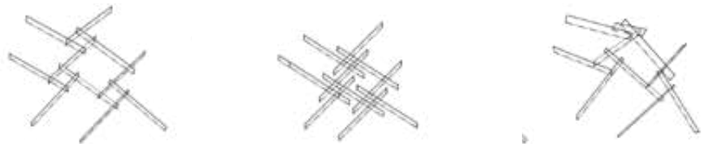
The design phase defines the geometric and structural logic of reciprocal shell systems through parametric modeling and form-finding. Design decisions consider assembly constraints, preparing the geometry for executable construction sequences.

Reciprocal Frame Research

Unit Generation



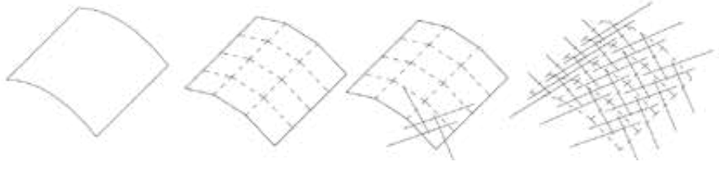
Form Variations



Parametric Modeling

Form Building

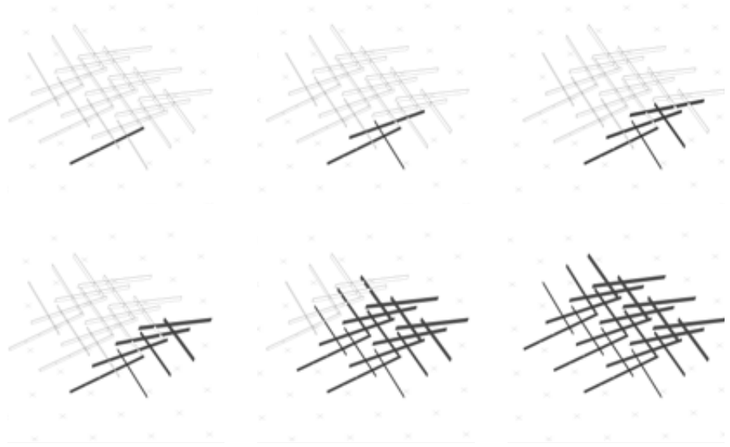
Module Generation



Learning

This stage bridges design and construction by formalizing assembly logic into learnable and programmable action sequences. Through sequence exploration and task learning, geometric intent is translated into executable robotic behavior.

Experiment of Construction Sequence



Programmable Sequence

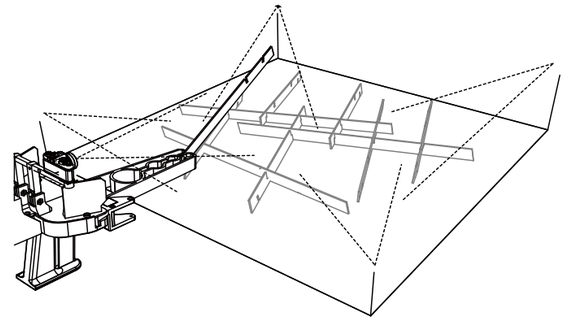
Atomic Actions

Sequence Composition

Loop Execution

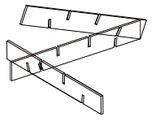
Task Learning

Environment Setting

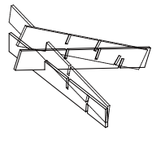


Disassemble Movements

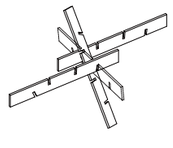
Movement A



Movement B



Movement C

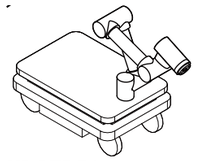


Construction

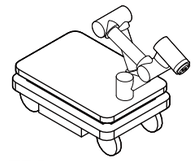
The construction phase translates programmed assembly logic into robotic execution through data-driven task learning and coordinated multi-robot operation. Sequences enable the system to autonomously perform reciprocal assembly under real conditions.

Hardware

Vehicle



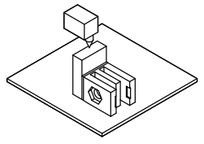
UR RTDE is used to control the robotic arms

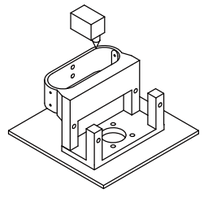


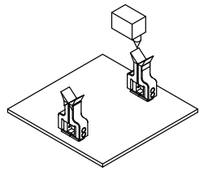
TF Radar is set on the bottom part to get real-time detection

Cameras

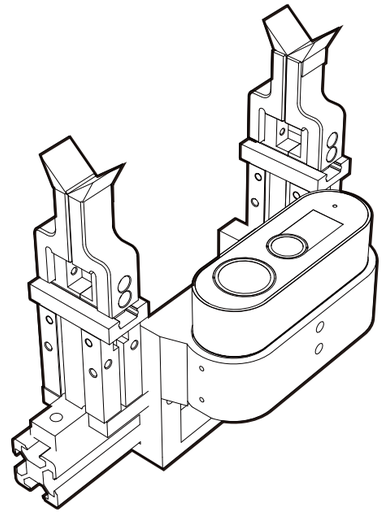
3D Printing







End Effector



Robotic System

Location

- SLAM
- Point Cloud

Component Positioning

- QR Tag
- Visual Guidance

Control System

- UR RTDE
- Motion Planning

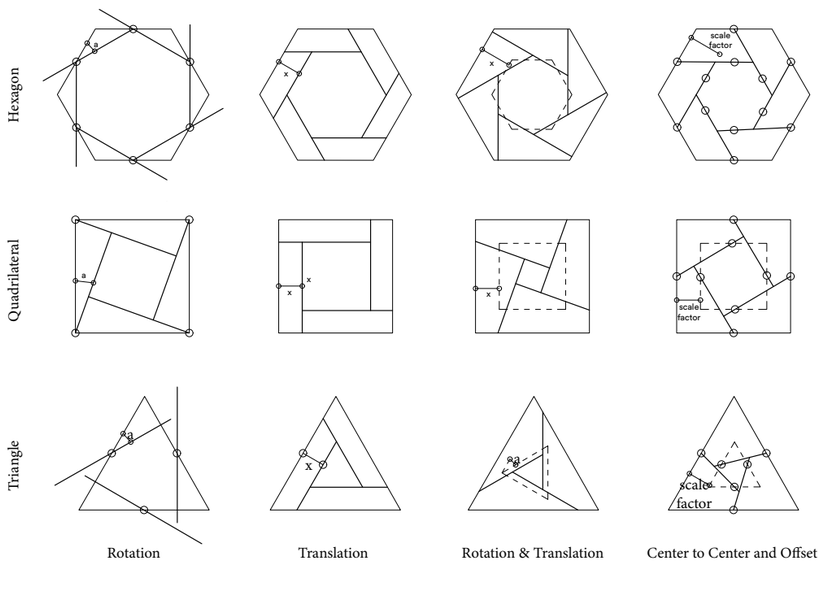
Parametric Design | *Parametric Strategies for Generating Reciprocal Shell Geometries*

This section demonstrates a parametric workflow for designing reciprocal structures. By establishing algorithmic logic, we drive the dynamic evolution of structural units while systematically analyzing how parametric variations influence global morphology. This digital methodology integrates design intent with geometric constraints, enabling the organic transformation of local rules into holistic spatial configurations. The process culminates in the generation of structures that harmonize structural rationality with expressive visual language.

Reciprocal Frame Research

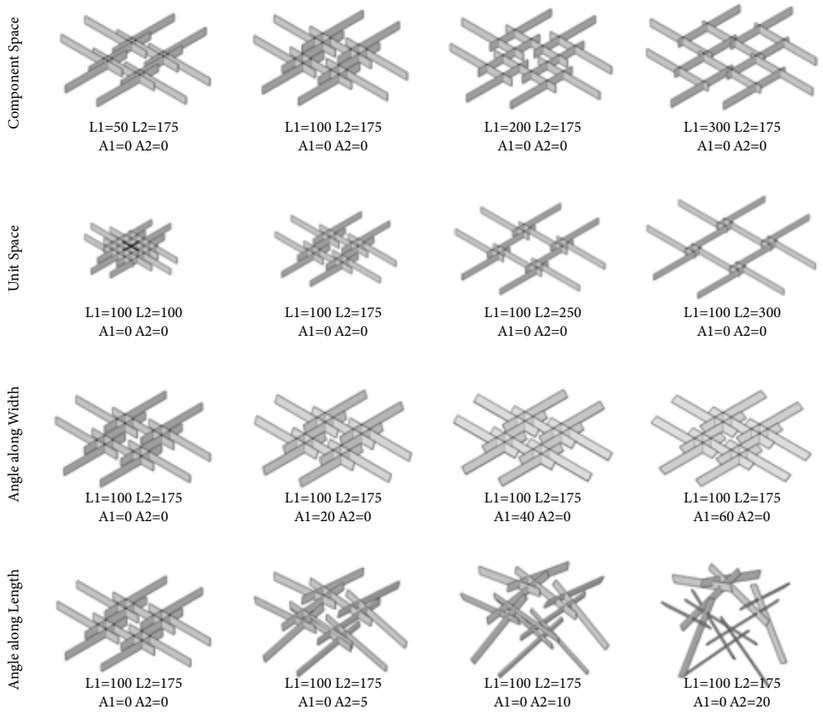
Unit Generation

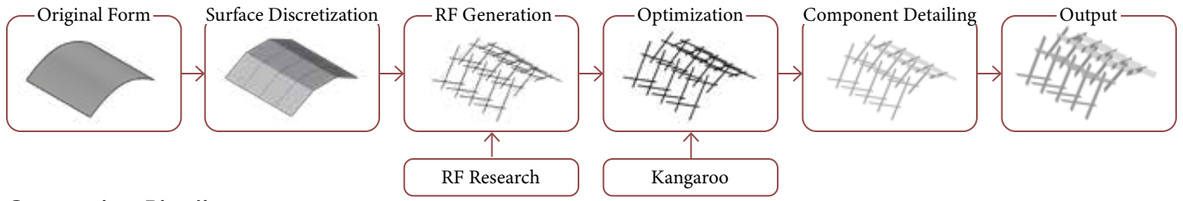
The generation of Reciprocal Frame units employs geometric transformations of base polygons. Key methods include rotation, translation, and the center-to-center approach. The latter creates RF units by scaling polygons, connecting midpoints of original and scaled shapes to intersection points, and offsetting these points to form eccentric connections.



Form Variations

This step then investigates the morphological relationships by tuning two key groups of parameters: the unit-level parameters, including member spacing (L1) and inclination angles along width (A1) and length (A2); and the system-level parameter of inter-unit distance (L2). This systematic variation allows for the exploration of a design space derived from a single reciprocal unit.

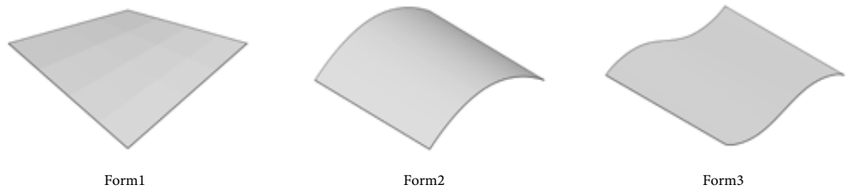




Generation Pipeline

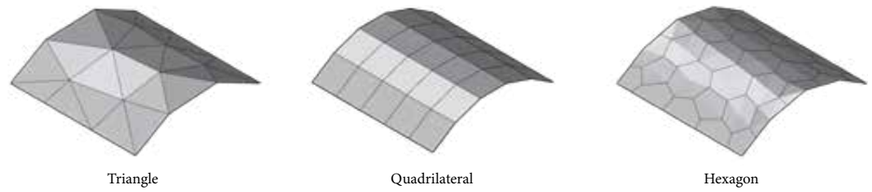
Original Form

The process begins with an arbitrary surface that serves as the geometric basis for all subsequent generation.



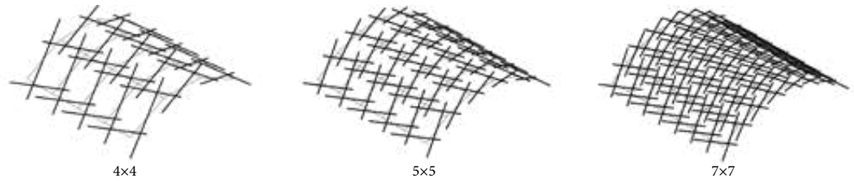
Surface Discretization

The surface is remeshed into triangular, quadrilateral, or hexagonal units to create a discrete geometric base for structural generation.



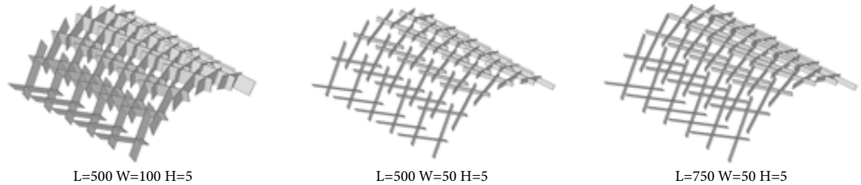
RF Generation

The pre-defined reciprocal unit types are applied to the discrete mesh, generating a spatially performative structural system.



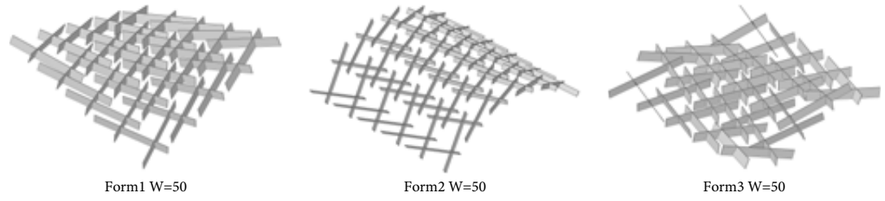
Component Detailing

This step involves refining panel thickness and notch dimensions, finalizing the component-level detailing to prepare for physical cutting and assembly.



Final Results

The final results show multiple reciprocal shell outcomes generated from a single parametric workflow, demonstrating formal variation under consistent structural logic.



Assembly Logic | *Deriving Programmable Sequences from Manual Construction Trials*

This page documents the exploration of assembly order for reciprocal units, aiming to identify a repeatable and programmable logic for robotic fabrication. Through iterative manual construction and robotic gripper simulations, the complex interlocking behavior was decomposed into a set of essential actions such as insertion, lifting, and bypassing. These distilled action patterns established the foundational action sequence for later automation, forming the rule-based framework that robots would eventually learn and execute.

Construction Experiment

Sequence 1
Pushing the lower plank to the slot of the upper plank

Sequence 2
Lifting the lower plank

Repeating Sequence 1
Stretch in the horizontal direction

Repeating Sequence 1
Stretch in the horizontal direction

Sequence 3
Bypassing the new plank from below to the slot of the original plank

Repeating Sequence 3
Stretch in the longitudinal direction

Repeating Sequence 3
Stretch in the longitudinal direction

Repeating Sequence 1 & 2
Stretch in the longitudinal direction

Repeating the Loop
Expand in the process of repetition

Repeating the Loop
Expand in the process of repetition

Repeating the Loop
Expand in the process of repetition

Repeating the Loop
Expand in the process of repetition

Repeating the Loop
Expand in the process of repetition

Repeating the Loop
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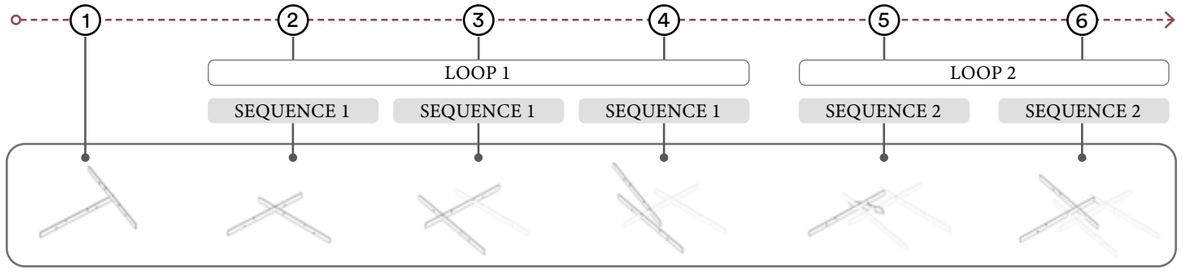
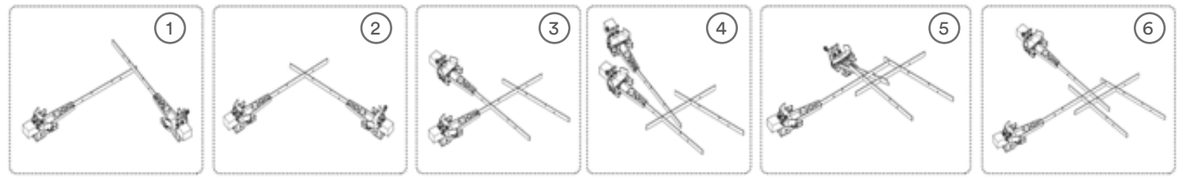
Repeating the Loop
Expand in the process of repetition

Repeating the Loop
Expand in the process of repetition

Programmable Sequence



Simulation



Task Decomposing

The assembly process was distilled into three essential action units—insertion, lifting, and bypass—capturing the fundamental mechanics of reciprocal interlocking. These actions formalize the otherwise complex manual operation into a clear, repeatable, and programmable sequence, establishing the behavioral rules that later guide robotic execution.

Sequence 1

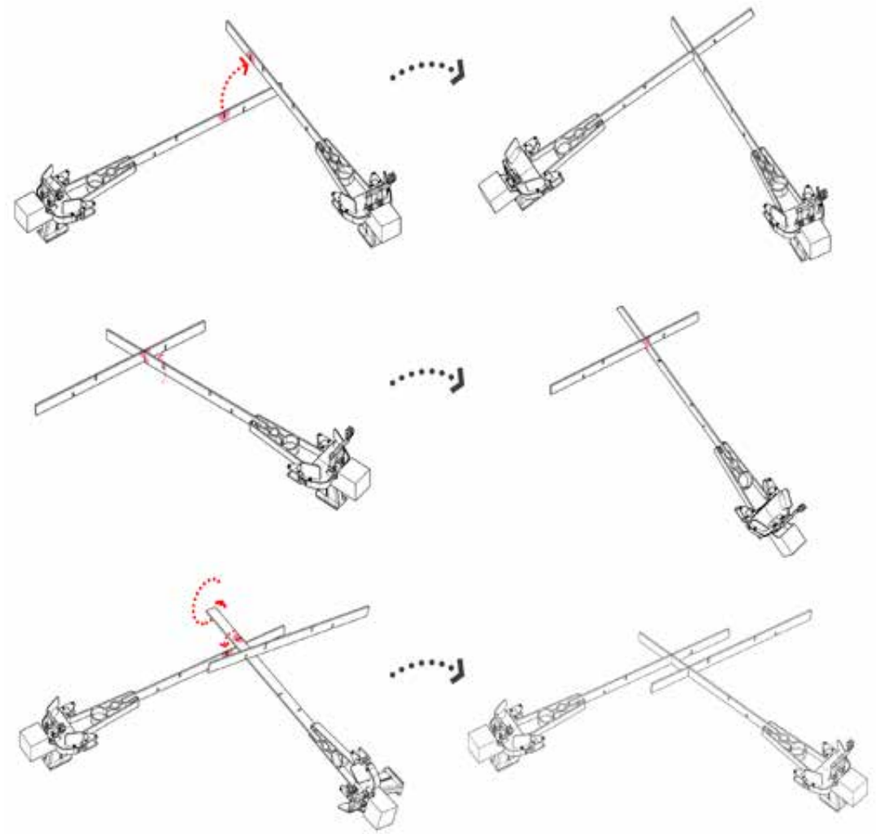
Pushing the lower plank to the slot of the upper plank

Sequence 2

Pushing the upper plank to the slot of the lower plank

Sequence 3

Bypassing the new plank from below to the slot of the original plank

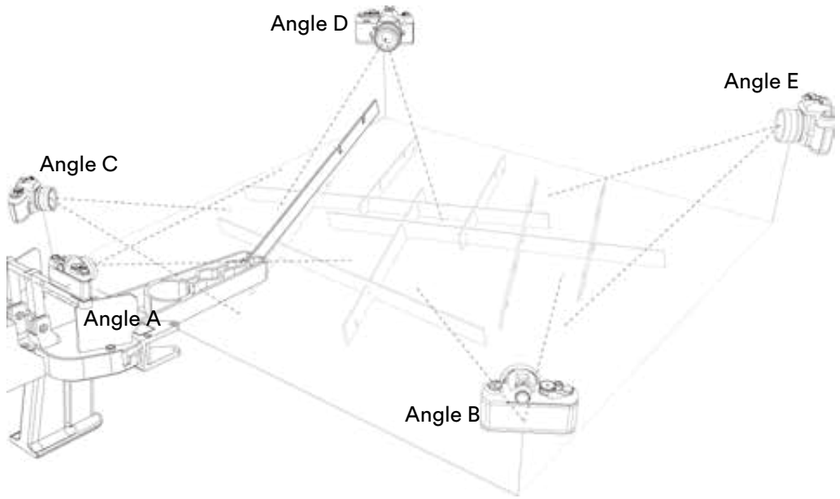


Task Learning | *Multi-View Capture and Network Training for Robotic Assembly*

This page presents the data collection and learning pipeline used to train the robot to reproduce the programmed assembly actions. A controlled environment with multi-view cameras and QR-based spatial tracking was built to capture precise action demonstrations. The collected video frames were labeled to train an action classification network, enabling the robot to recognize and sequence the learned behaviors. The resulting model successfully reproduces the full assembly routine, validating the data-driven approach to reciprocal unit construction.

Data Collection

Data was collected through multi-angle video capture and QR-code-aided positioning to train the robotic task learning model.



Perspective A

Perspective B

Perspective C

Perspective D

Perspective E

Top Perspective

Video Frame Extraction

Timeline

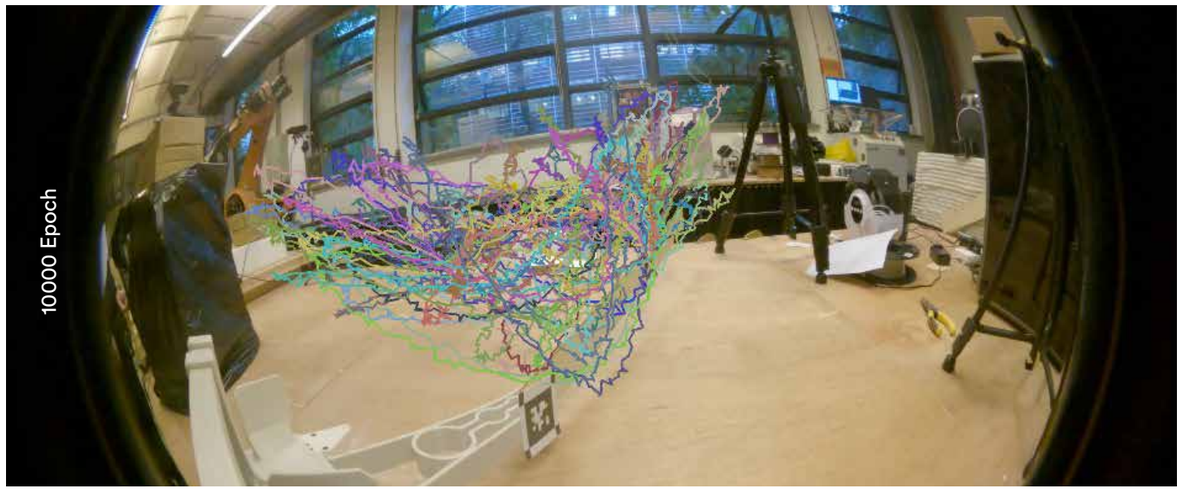
Sequence 1
(Cam C, Cam D, Cam B, Cam E)

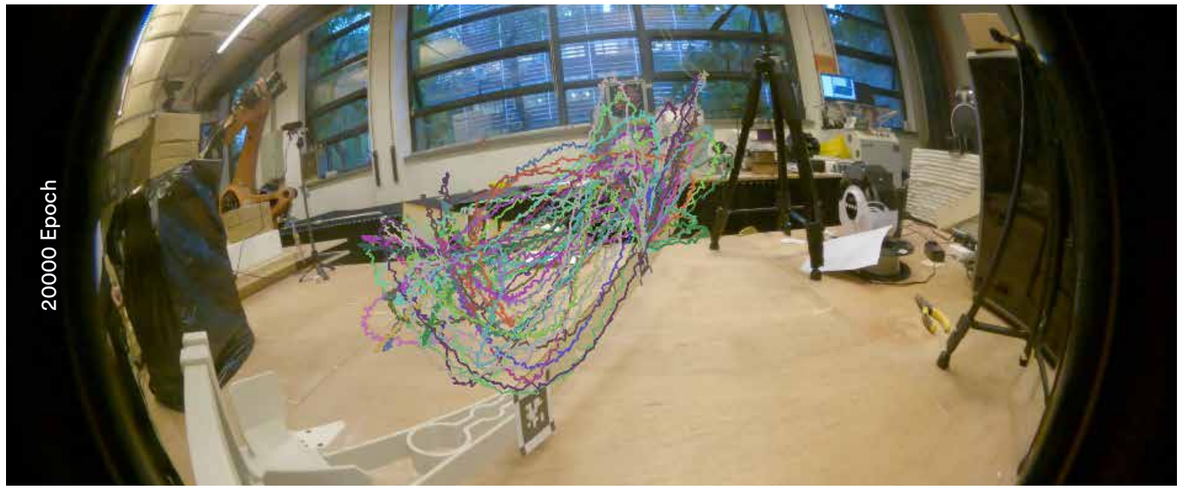
Sequence 2
(Cam B, Cam C, Cam D, Cam E)

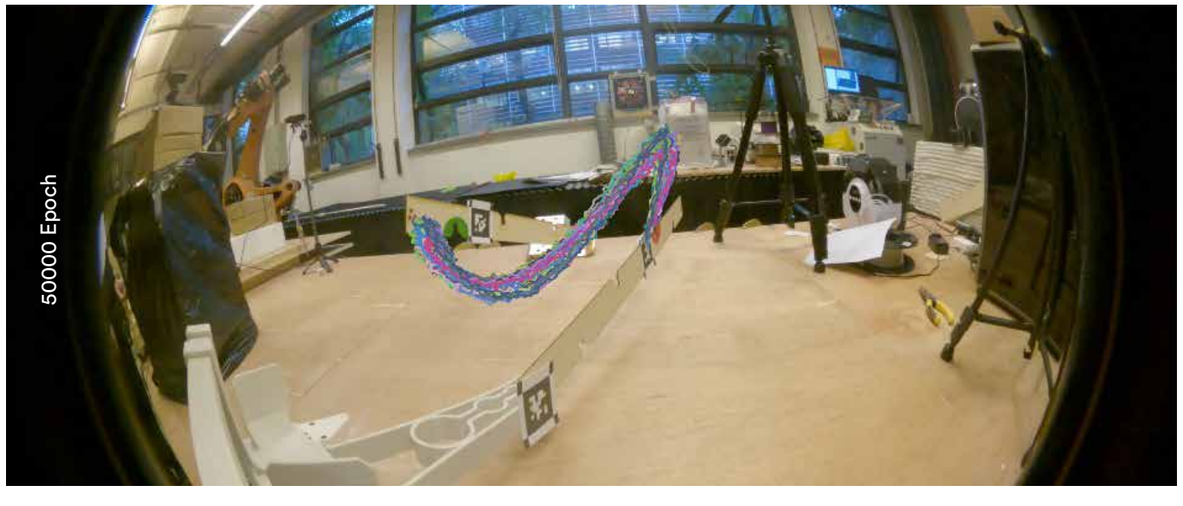
Sequence 3
(Cam B, Cam C, Cam D, Cam E)

Outcome

Through multi-angle video and QR-code positioning data, the robotic arm has successfully learned and mastered the complete action sequence for reciprocal unit insertion.





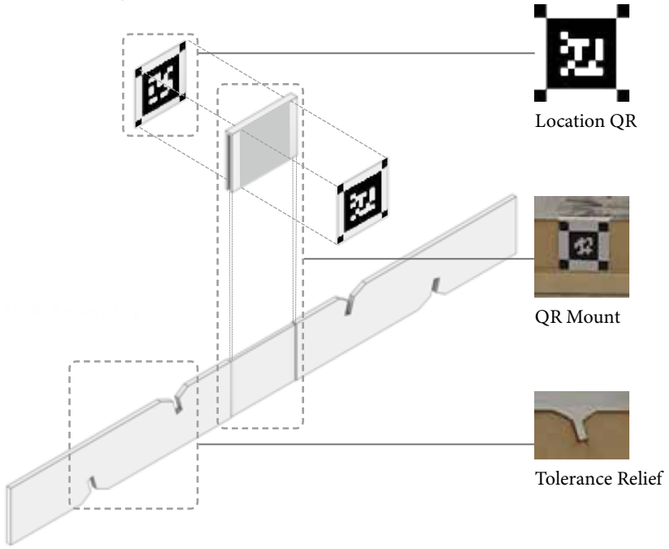


Construction Details | *Detailing components, End-effectors, and System integration*

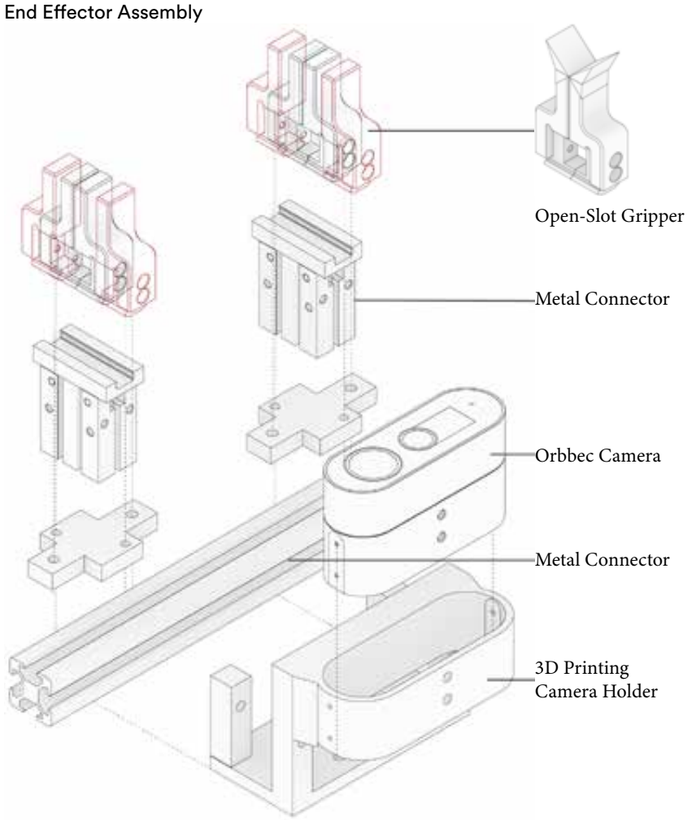
This section delves into the physical embodiment of the reciprocal structure, dissecting its integrated hardware systems. It visually articulates the complete assembly, from customized building units and a sensor-laden mobile platform to bespoke end-effectors. Individual motion tests were conducted to validate the precision of each component and the synergy of the integrated system.

Details

Unit Assembly

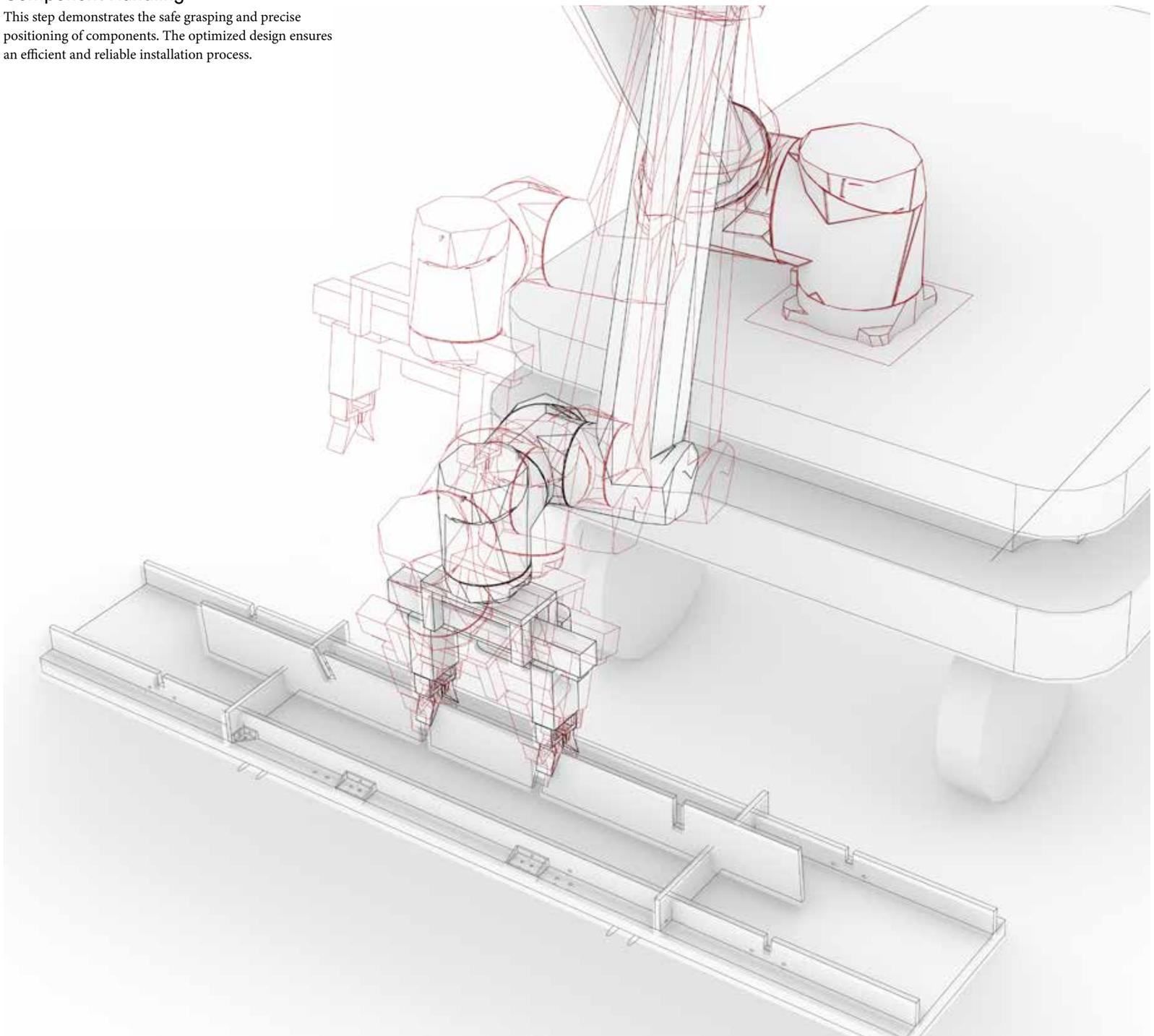


End Effector Assembly



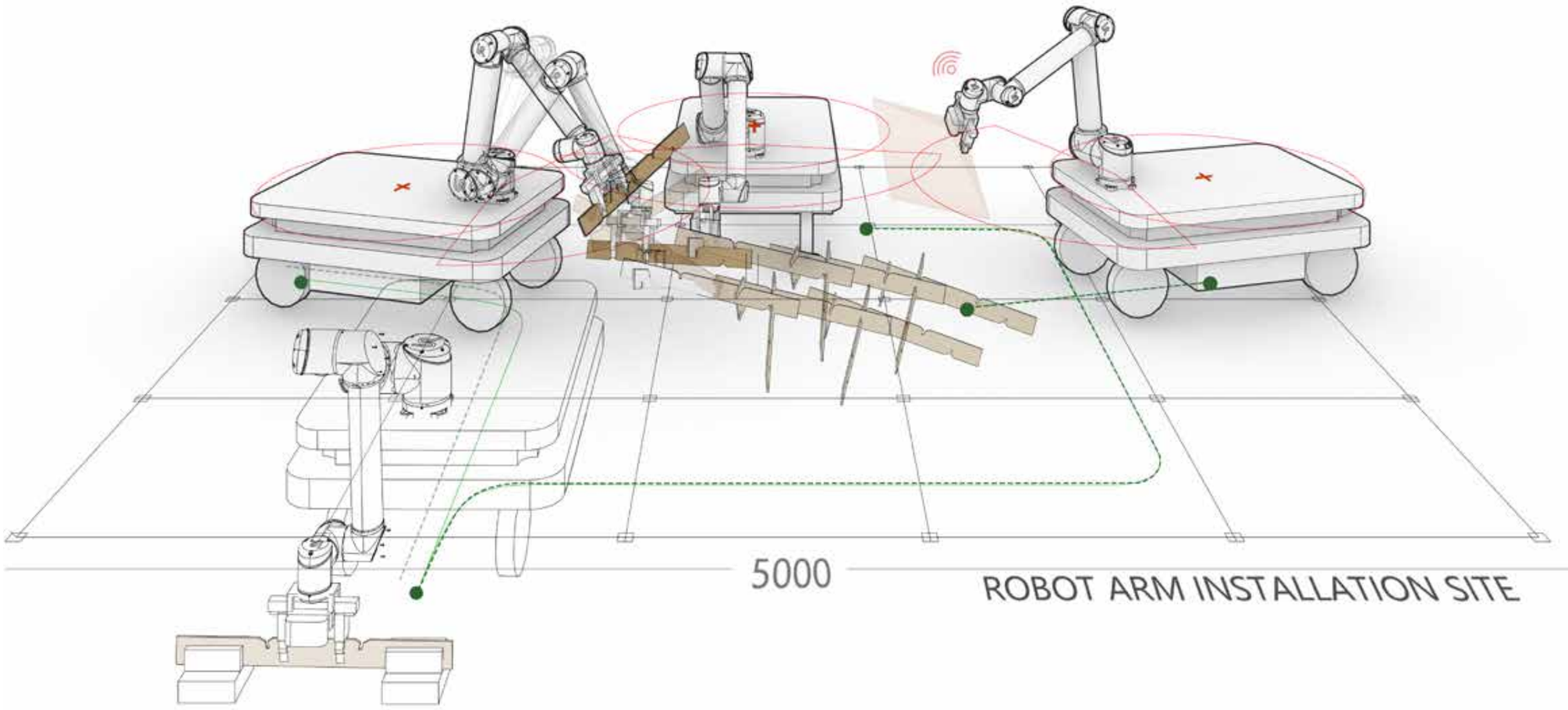
Component Handling

This step demonstrates the safe grasping and precise positioning of components. The optimized design ensures an efficient and reliable installation process.



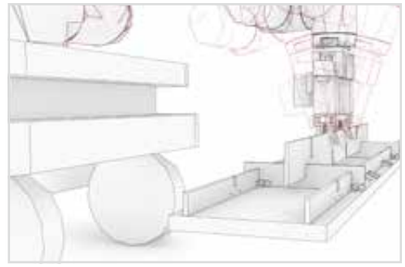
Multi-Robots Collaboration | *Coordinating Perception and Action across Multiple Robots*

This page showcases the synchronized operation of a multi-robot system in a collaborative fabrication workflow. The process initiates with a robotic arm precisely acquiring a building component, followed by an agile mobile platform navigating to the designated location via LiDAR positioning. The system demonstrates coordinated collaboration, with one unit providing visual guidance while others perform physical assembly. However, current limitations include a lack of fault tolerance—the system halts if any component placement fails—and increasing structural loads during later assembly stages, which impose higher demands on robotic strength. Through this process, the systemic integration of three robots highlights the importance of role allocation and workflow coordination in collective robotic construction.



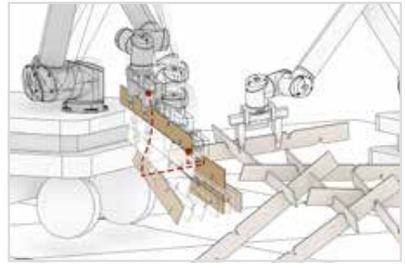
Atomic Action

Material Handling



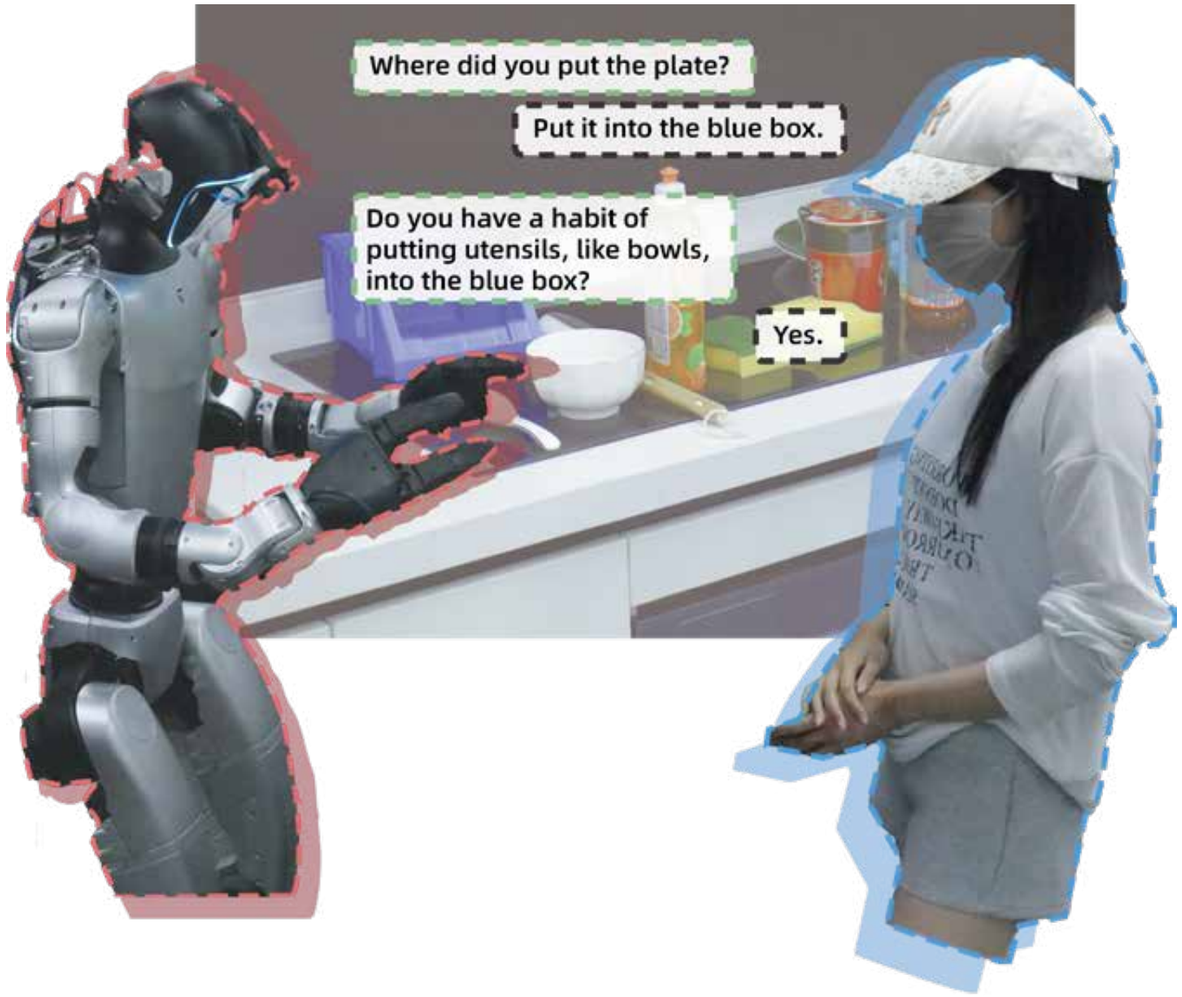
Part Positioning

Task Execution



Assembly in Progress

These site photographs document the translation from digital models to physical construction. They validate the design intent while capturing the precision and collaboration inherent in the assembly process.



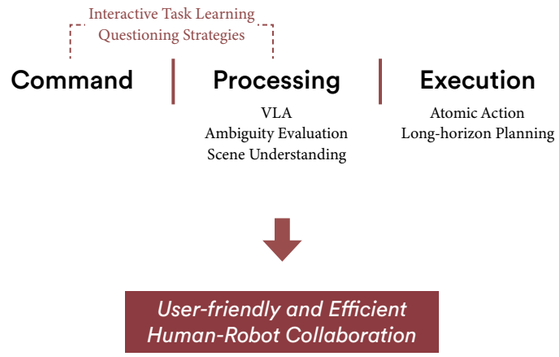
Adaptive Querying

Questioning Strategies for Robot Learning in Household Tasks

Professional Project

This project investigates how robots can learn to perform long-horizon household tasks through structured questioning that uncovers user preferences and resolves ambiguity. It introduces a configurable vision-language–based questioning framework that operationalizes human-derived inquiry patterns as controllable robotic strategies. The pipeline integrates scene understanding, structured preference parsing, and strategy-driven question generation. Through deployment on a humanoid robot and controlled user studies, the system demonstrates that different questioning strategies lead to stable interaction behaviors and significantly distinct user perceptions. The project establishes questioning strategy as a designable computational primitive for human–robot interaction, enabling systematic exploration of preference learning and interaction quality in real-world tasks. I am motivated to explore working methods that enable humans and robots to collaborate more efficiently and comfortably through well-designed interaction structures.I hope to explore workflow-driven approaches in which shell design, geometric optimization, and construction are treated as an interconnected and mutually influential process.

Keywords: Human-Robot Interaction, Questioning Strategies, Vision-Language Models, User Study
Instructor: Weiwei Guo, Xiaohua Sun
Role: VLA Study, HRI Experiment Design, User Study
Team: Suming Li, Yuanda Hu, Dai Shi, Jiani Hou, Yujie Qiu
Contribution: Formative Study, Experiment Design, User Study
Year: 2025



Interaction Strategies | *Research on questioning-driven robot learning*

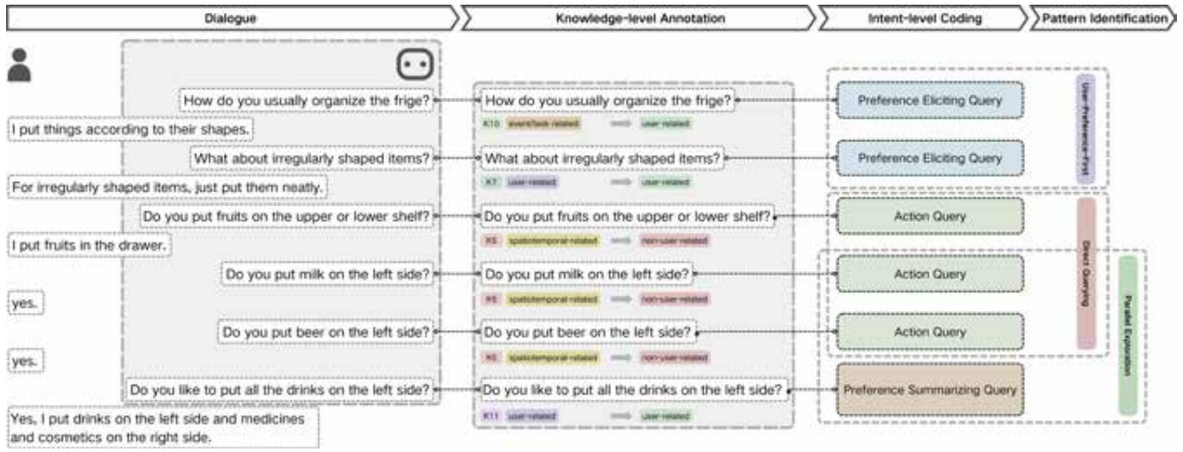
This project reframes questioning as a designable interaction strategy in human–robot research. By abstracting human teaching behaviors into structured questioning patterns, it builds a system that allows robots to learn tasks through different interaction logics. The work focuses on how strategy-level differences influence learning dynamics and user perception, rather than optimizing individual questions.

Illustration of the experimental procedure

Left: In the refrigerator organization task, the learner asks about the teacher's organizing principles.
Right: In the cocktail mixing task, the learner queries the teacher's preferences.

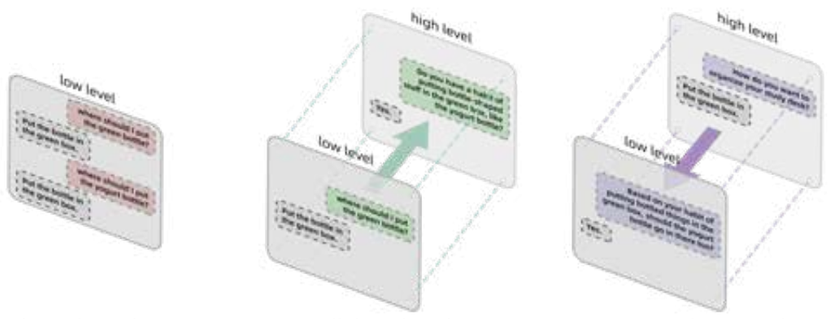
Illustration of the annotation pipeline

Starting from raw human–robot dialogues, each question is first labeled by knowledge type, then coded for interactional intent, and finally abstracted into higher-level questioning patterns. This hierarchical process bridges low-level utterances and dialogue-level strategies.



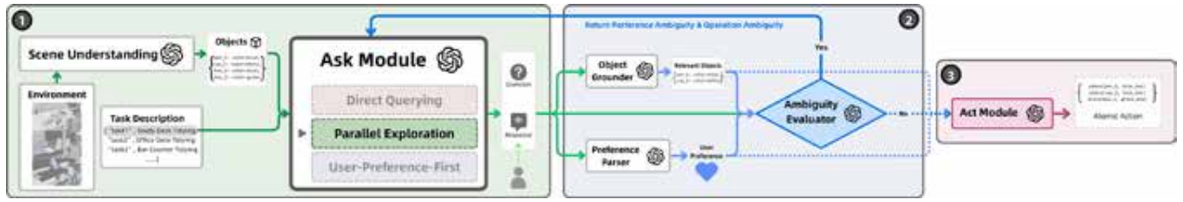
Questioning Pattern

This diagram illustrates the three core questioning strategies derived from human teaching patterns. It contrasts their distinct approaches: from the reactive, item-by-item Direct Querying, to the top-down, rule-first User-Preference-First, and the adaptive, interleaved Parallel Exploration, showcasing a spectrum of human-to-robot information gathering.

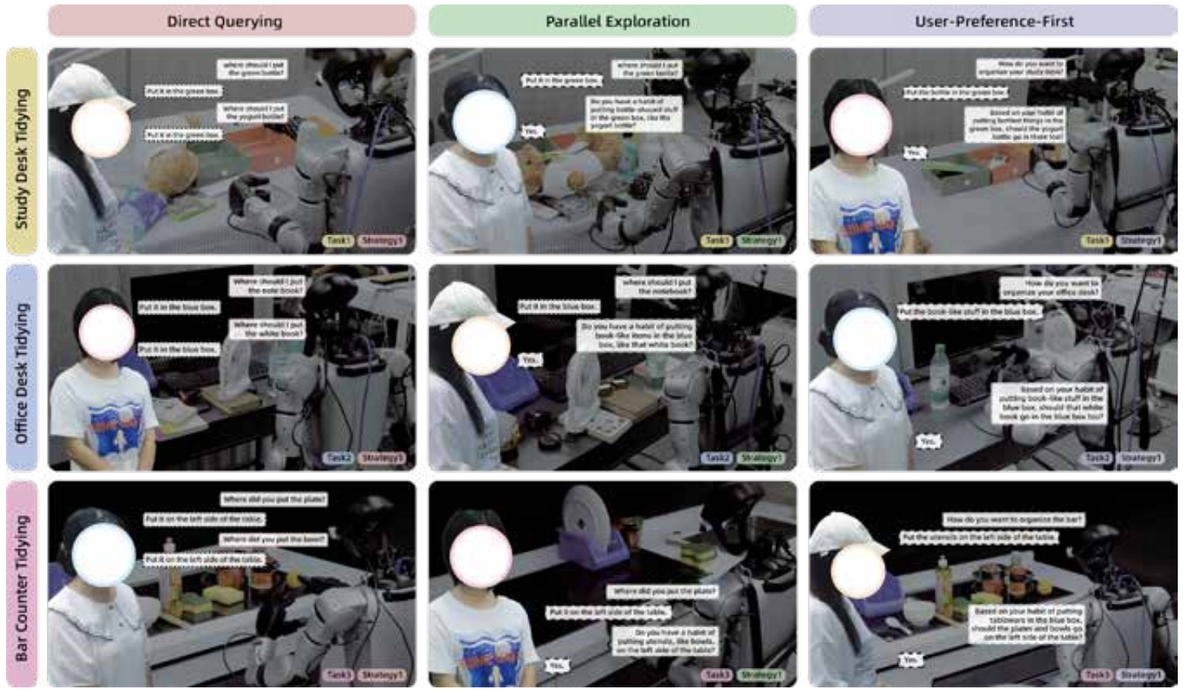


Direct Querying | Parallel Exploration | User-Preference-First

System Architecture



Experiment



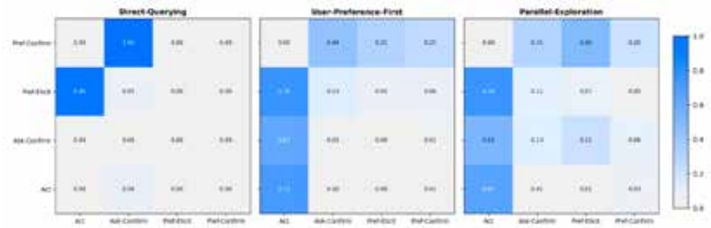
Conclusion

User Study Findings

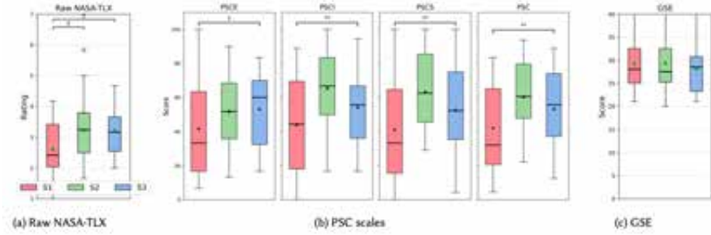
- Each strategy yielded perceptibly different interaction experiences and user perceptions.
- Parallel Exploration was favored for its intelligence and efficiency, despite inducing higher cognitive load.
- Direct Querying was perceived as repetitive.
- User-Preference-First was viewed as accurate but slower.
- The study reveals a key trade-off between user experience and interaction efficiency.

Implications & Future Directions

- Provides a reproducible framework for instantiating and comparing questioning strategies.
- Paves the way for adaptive robots capable of dynamically switching strategies based on user and context, enabling seamless long-term collaboration.



Whether human-derived patterns can be deployed as stable strategies on robots



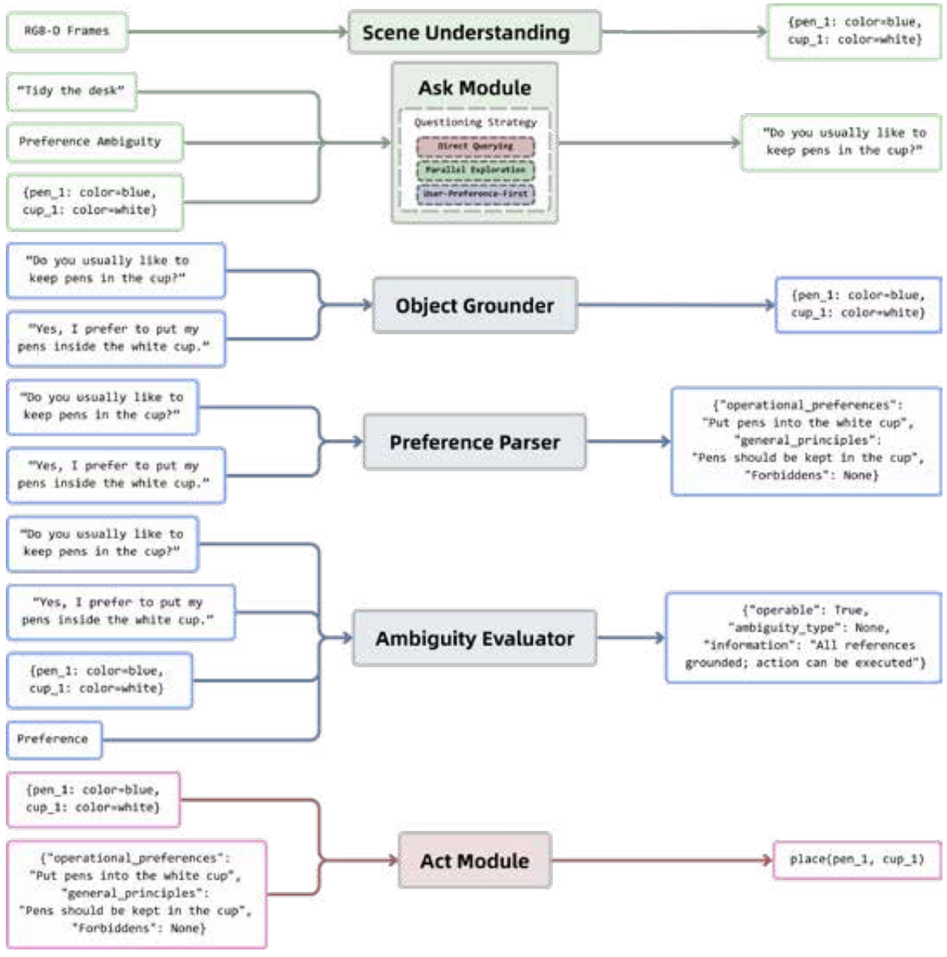
How do different strategies influence user workload and interaction experience?

System Implementation | *Framework Design & User Study*

We developed a modular VLM-based framework that instantiates three questioning strategies on a humanoid robot. Through a user study (N=18) across three tidying scenarios, we evaluated strategy differences in workload, curiosity, and self-efficacy, supported by quantitative interaction logs and qualitative feedback.

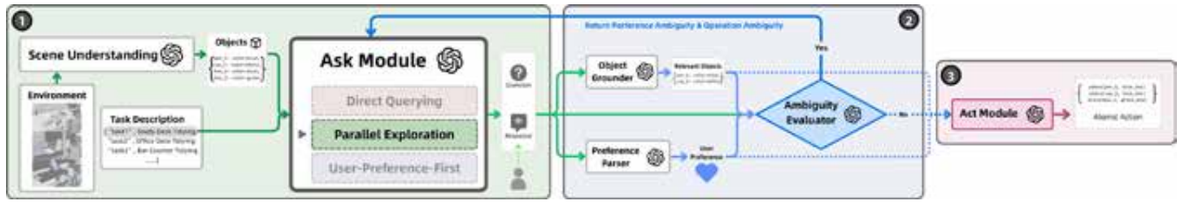
System Architecture

Overview of our system architecture. (1) The Scene Understanding module interprets the environment and task description to extract objects. (2) The Ask Module deploys questioning strategies (Direct Querying, Parallel Exploration, User-Preference-First) and interfaces with ambiguity evaluation through object grounding and preference parsing. (3) The Act Module executes atomic actions once ambiguities are resolved.



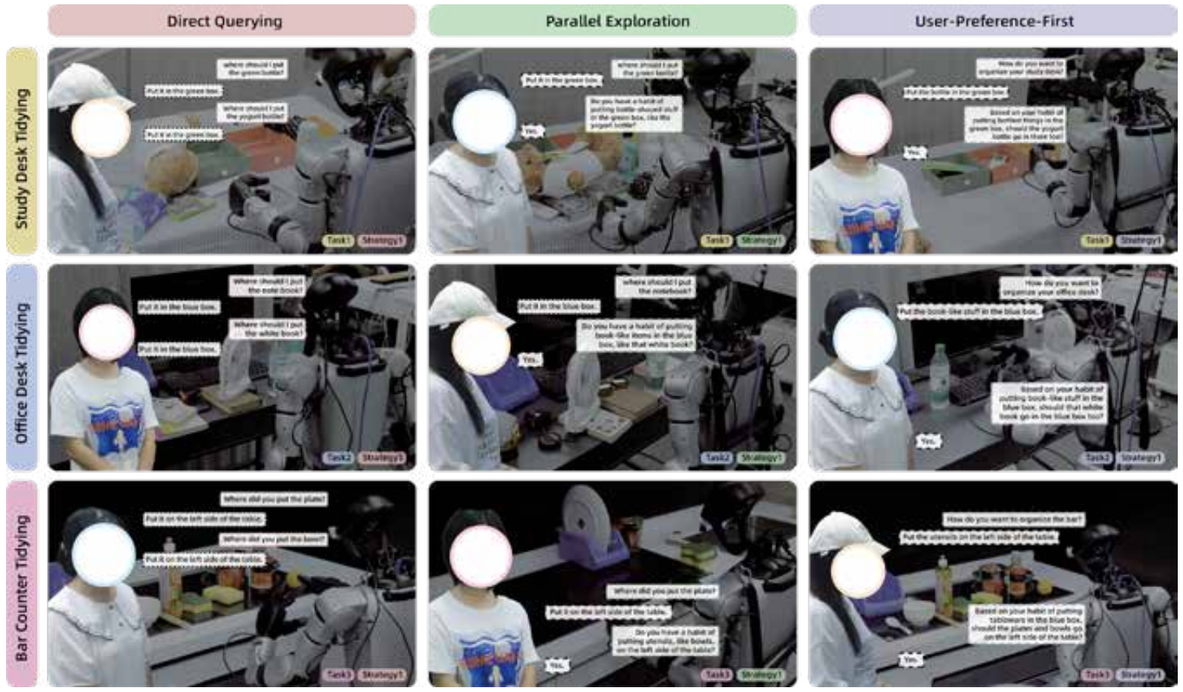
Experiment Setting

Initial layouts for the three household tidying tasks: bar counter (left), study desk (middle), and office desk (right). Each scenario contains approximately 10–12 items and 2–3 storage spaces, designed with comparable complexity while preserving room for individual preferences.



Experiment

Illustration of the three questioning strategies (Direct Querying, Parallel Exploration, User-Preference-First) across three task scenarios (study desk tidying, office desk tidying, bar counter tidying).



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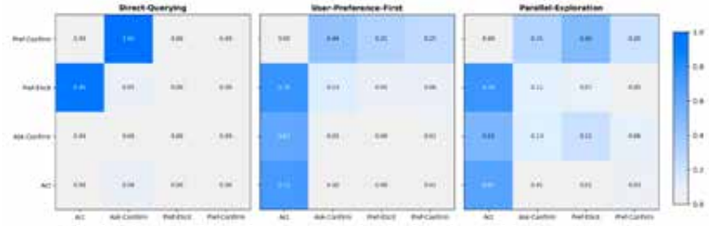
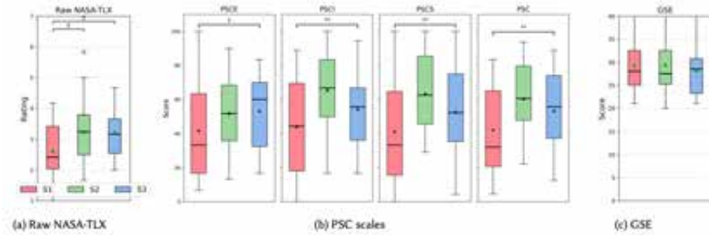


Whether human-derived patterns can be deployed as stable strategies on robots



How do different strategies influence user workload and interaction experience?